

ECE 3317

Applied Electromagnetic Waves

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Notes 18

Reflection and Transmission of Plane Waves



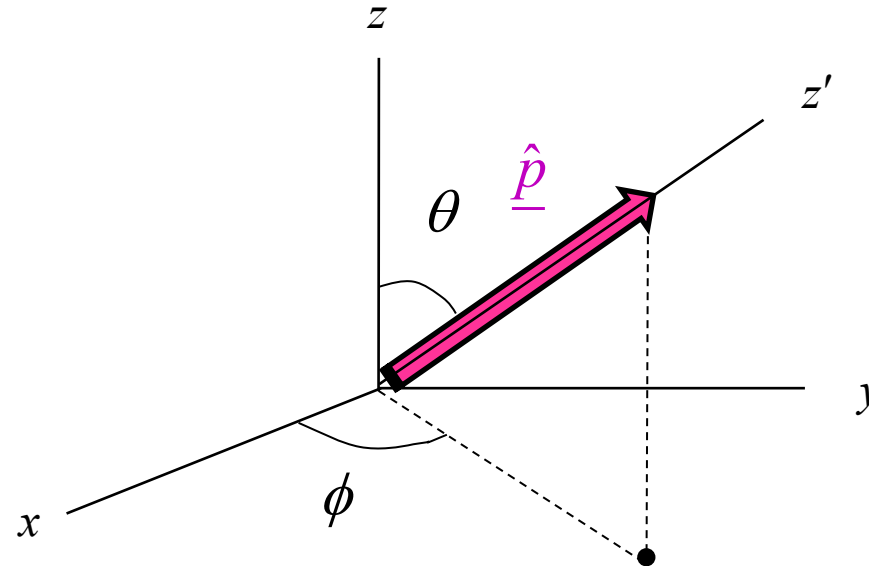
General Plane Wave

Consider a plane wave traveling in an arbitrary direction $\underline{\hat{p}}$ in space.

Denote:

$$\psi = e^{-jkz'}$$

(This function ψ tells us how the plane wave varies in space.)



We call the axis of propagation z' .

$$\underline{\hat{p}} = \underline{\hat{x}}(\sin \theta \cos \phi) + \underline{\hat{y}}(\sin \theta \sin \phi) + \underline{\hat{z}}(\cos \theta)$$

(This is the direction of power flow.)

$$z' = x(\underline{\hat{p}} \cdot \underline{\hat{x}}) + y(\underline{\hat{p}} \cdot \underline{\hat{y}}) + z(\underline{\hat{p}} \cdot \underline{\hat{z}})$$

We then have:

$$z' = x(\sin \theta \cos \phi) + y(\sin \theta \sin \phi) + z(\cos \theta)$$

General Plane Wave (cont.)

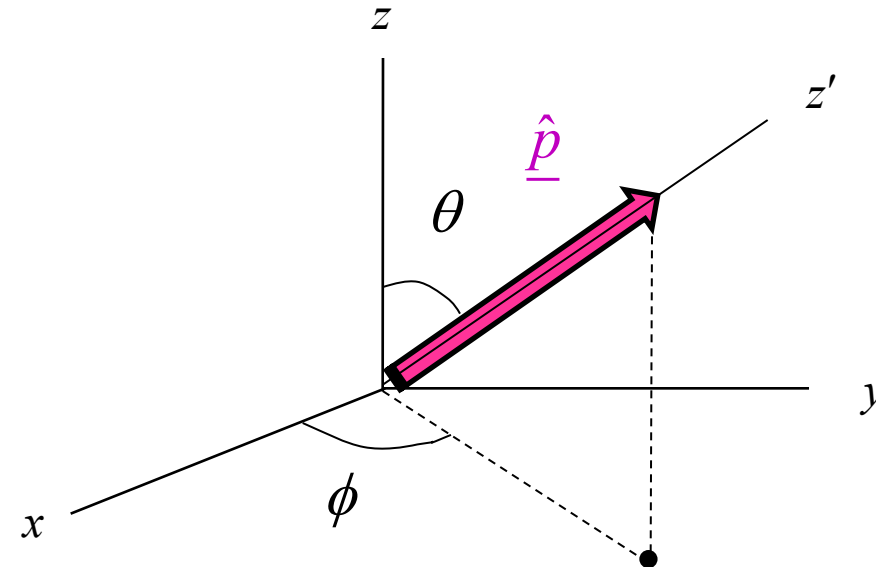
Hence

$$\psi = e^{jkz'} = e^{-j(k_x x + k_y y + k_z z)}$$

$$k_x = k \sin \theta \cos \phi$$

$$k_y = k \sin \theta \sin \phi$$

$$k_z = k \cos \theta$$



Note:

$$k_x^2 + k_y^2 + k_z^2 = k^2 \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + k^2 \cos^2 \theta$$

or

$$k_x^2 + k_y^2 + k_z^2 = k^2 \quad (\text{wavenumber equation})$$

General Plane Wave (cont.)

We define the **wavevector**:

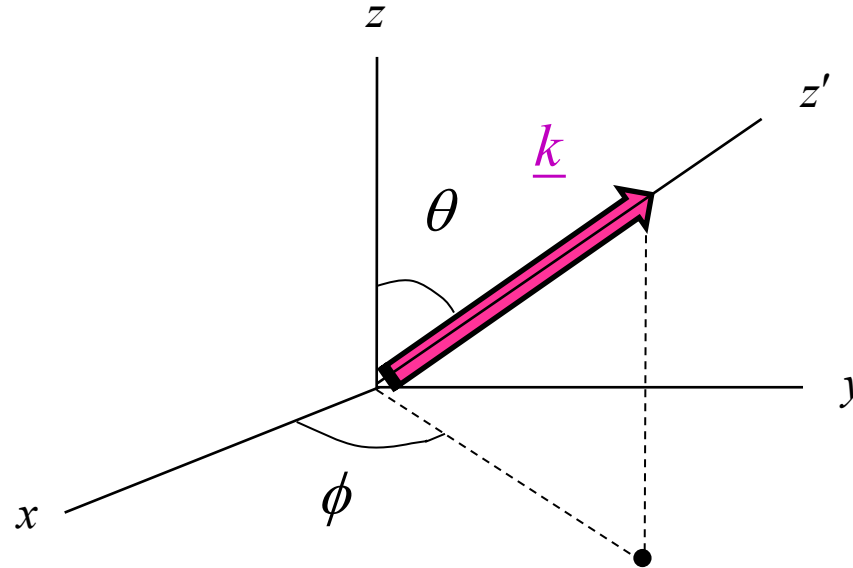
$$\underline{k} \equiv \underline{\hat{x}} k_x + \underline{\hat{y}} k_y + \underline{\hat{z}} k_z$$

$$k_x = k \sin \theta \cos \phi$$

$$k_y = k \sin \theta \sin \phi$$

$$k_z = k \cos \theta$$

$$\underline{k} = k \underline{\hat{p}}$$



The $\underline{k}$ vector tells us which direction the wave is traveling in.

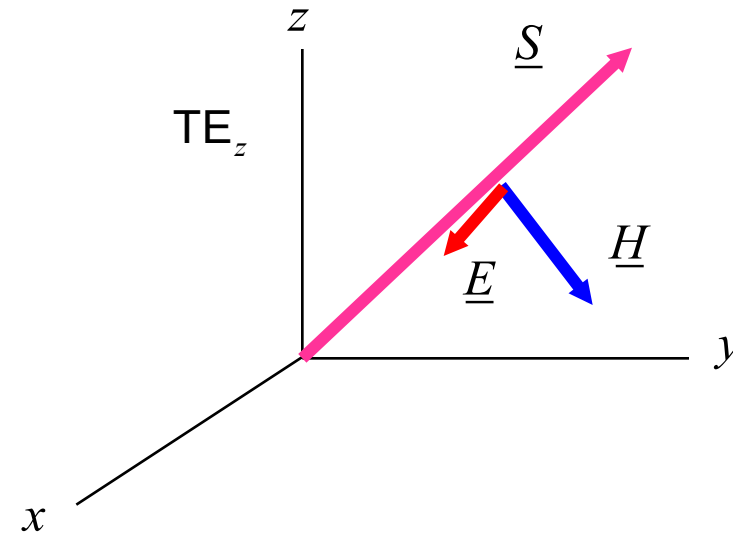
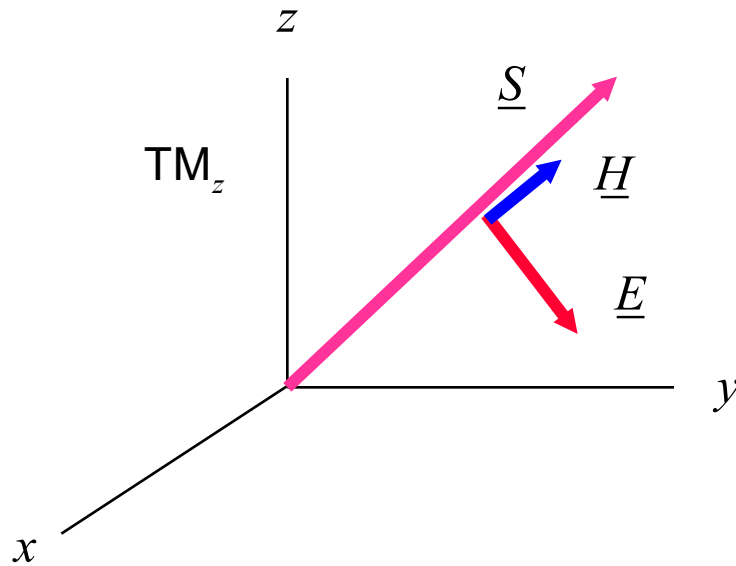
TM_z and TE_z Plane Waves

There are two fundamental polarizations possible.

- Transverse Magnetic (TM_z) $H_z = 0$
- Transverse Electric (TE_z) $E_z = 0$

$$\text{TM}_z: \underline{E} = \hat{\theta} E_\theta, \underline{H} = \hat{\phi} H_\phi$$

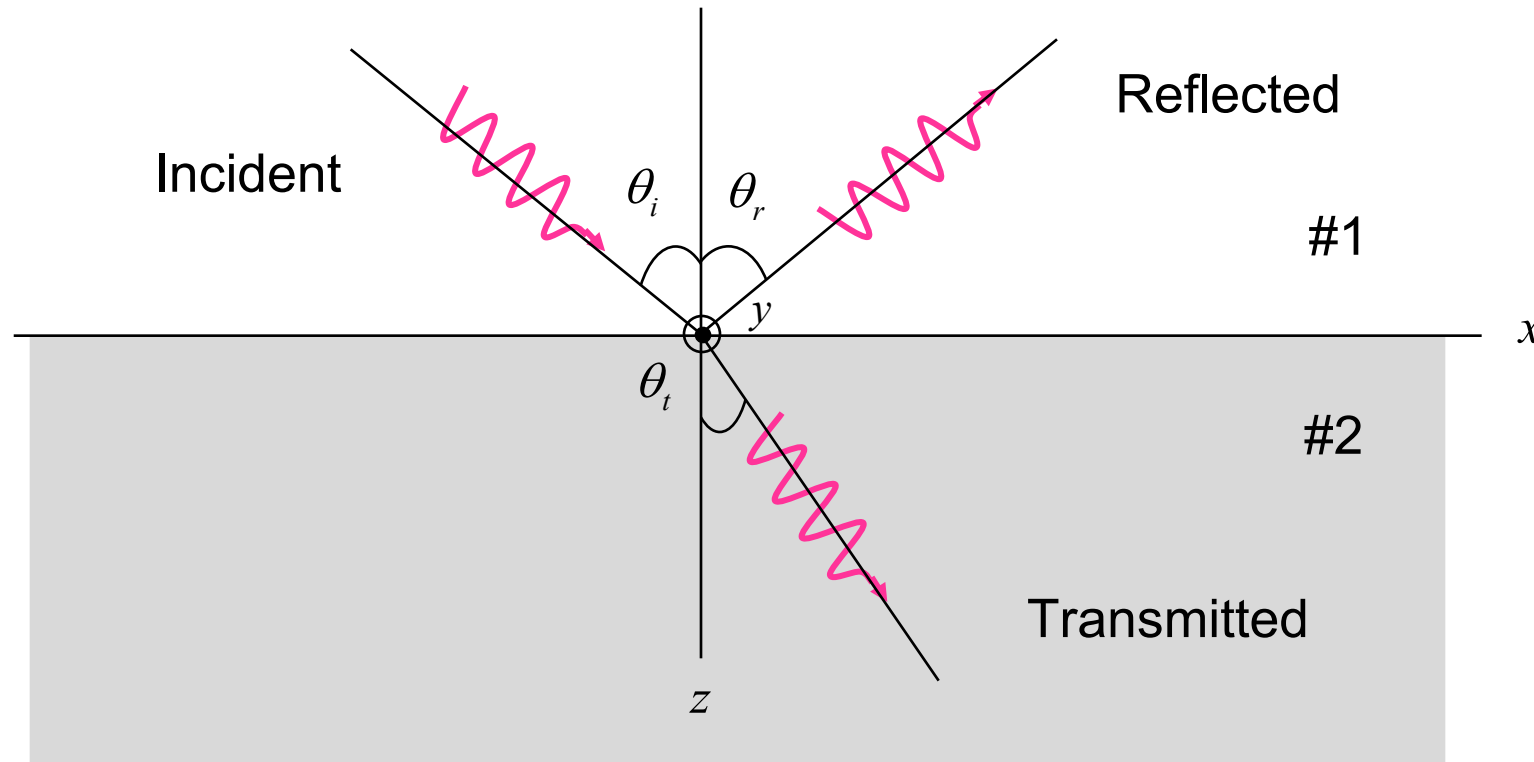
$$\text{TE}_z: \underline{E} = \hat{\phi} E_\phi, \underline{H} = \hat{\theta} H_\theta$$



Note: The word “transverse” means “perpendicular to.”

Reflection and Transmission

As we will show, each type of plane wave (TE_z and TM_z) reflects differently (and independently) from a material interface.



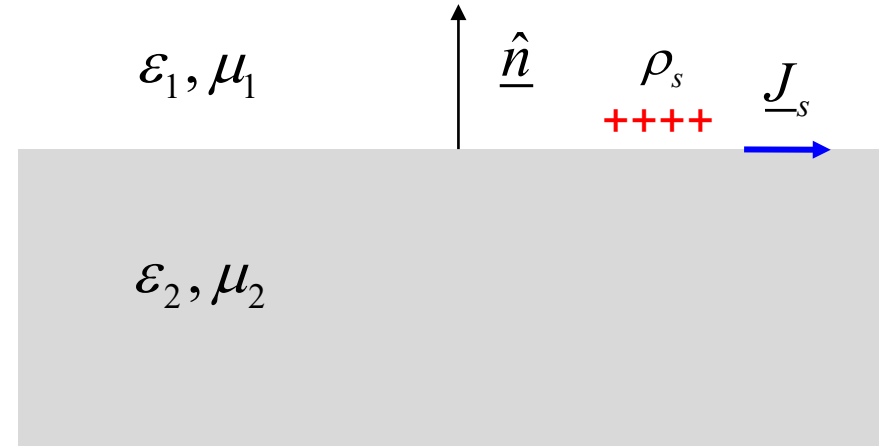
Note: We assume that the Poynting vector of the incident plane wave lies in the xz plane ($\phi = 0$). This is called the plane of incidence. There is then no variation of the fields in the y direction ($k_y = 0$).

Boundary Conditions

Here we review the boundary conditions at an interface (from ECE 3318).

General BCs:

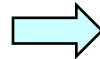
$$\begin{aligned}\hat{n} \cdot (\underline{D}_1 - \underline{D}_2) &= \rho_s \\ \hat{n} \times (\underline{E}_1 - \underline{E}_2) &= \underline{0} \\ \hat{n} \cdot (\underline{B}_1 - \underline{B}_2) &= 0 \\ \hat{n} \times (\underline{H}_1 - \underline{H}_2) &= \underline{J}_s\end{aligned}$$



Note: The unit normal points towards region 1.

No currents or charges on interface:

$$\begin{aligned}\hat{n} \times \underline{E}_1 &= \hat{n} \times \underline{E}_2 \\ \hat{n} \times \underline{H}_1 &= \hat{n} \times \underline{H}_2 \\ \hat{n} \cdot \underline{D}_1 &= \hat{n} \cdot \underline{D}_2 \\ \hat{n} \cdot \underline{B}_1 &= \hat{n} \cdot \underline{B}_2\end{aligned}$$

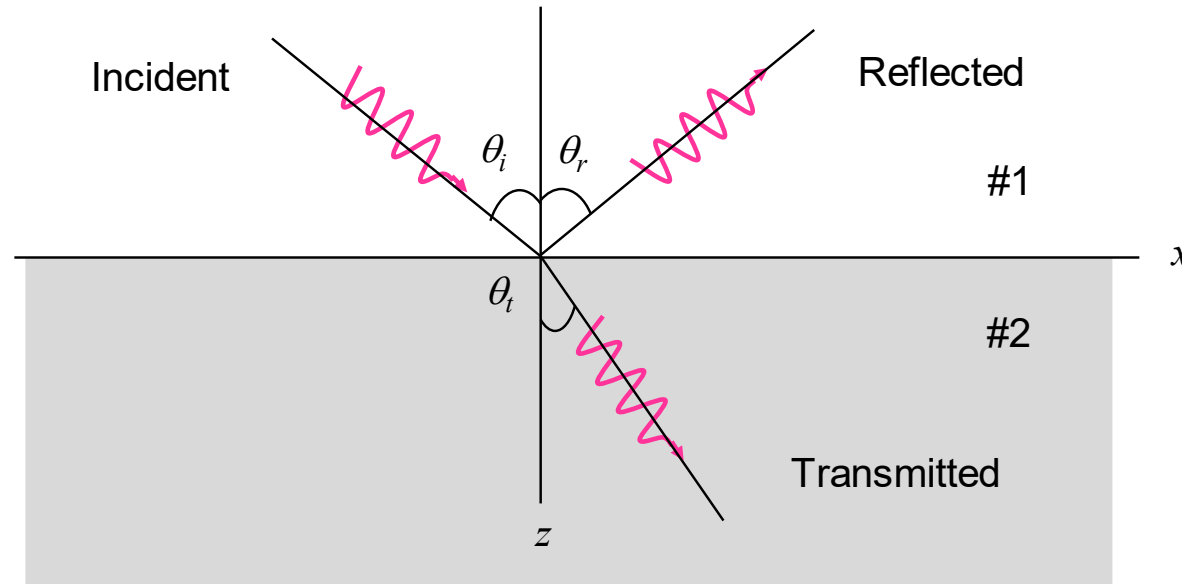


The tangential electric and magnetic fields are continuous.

(The normal electric and magnetic flux densities are also continuous.)

Reflection at Interface

Assume that the Poynting vector of the incident plane wave lies in the xz plane ($\phi = 0$). This is called the plane of incidence.



We first consider the (x,z) variation of the fields. (We will worry about the polarization later.)

$$\underline{E}^i = \underline{E}_{0i} e^{-jk_{xi}x - jk_{zi}z} \quad \underline{E}^r = \underline{E}_{0r} e^{-jk_{xr}x + jk_{zr}z} \quad \underline{E}^t = \underline{E}_{0t} e^{-jk_{xt}x - jk_{zt}z}$$

↑

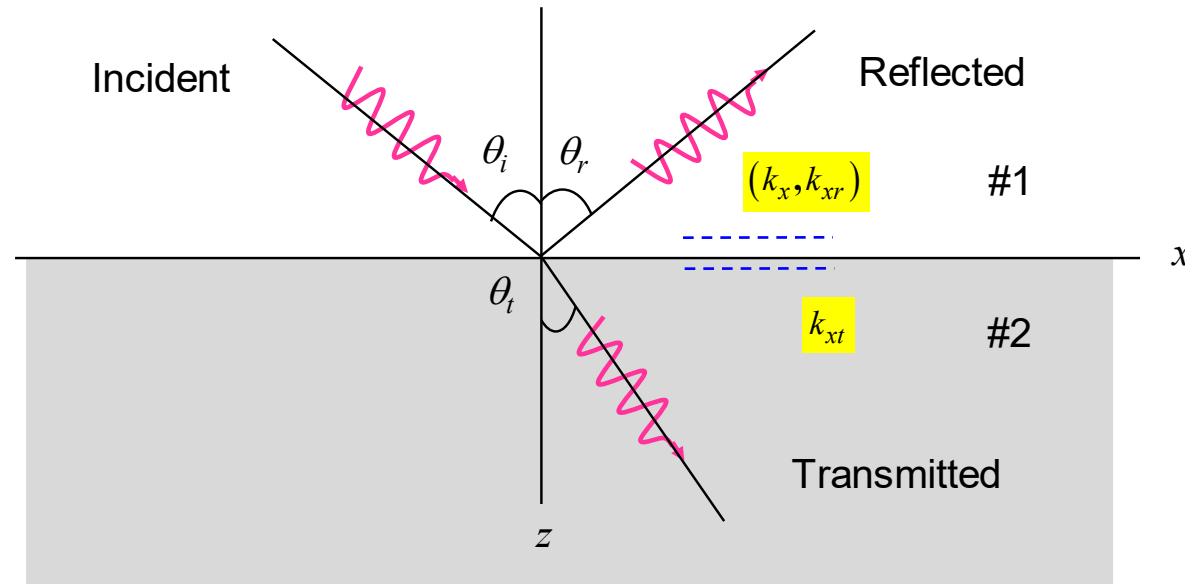
Note:

The plus sign for the exponent term in the reflected wave is chosen because this wave is traveling in the negative z direction.

Reflection at Interface (cont.)

Phase matching condition:

$$k_{xi} = k_{xr} = k_{xt}$$



This follows from the fact that the fields must match everywhere (for all x) at the interface ($z = 0$).

$$\underline{E}^i(x, 0) = \underline{E}_{0i} e^{-jk_{xi}x}$$

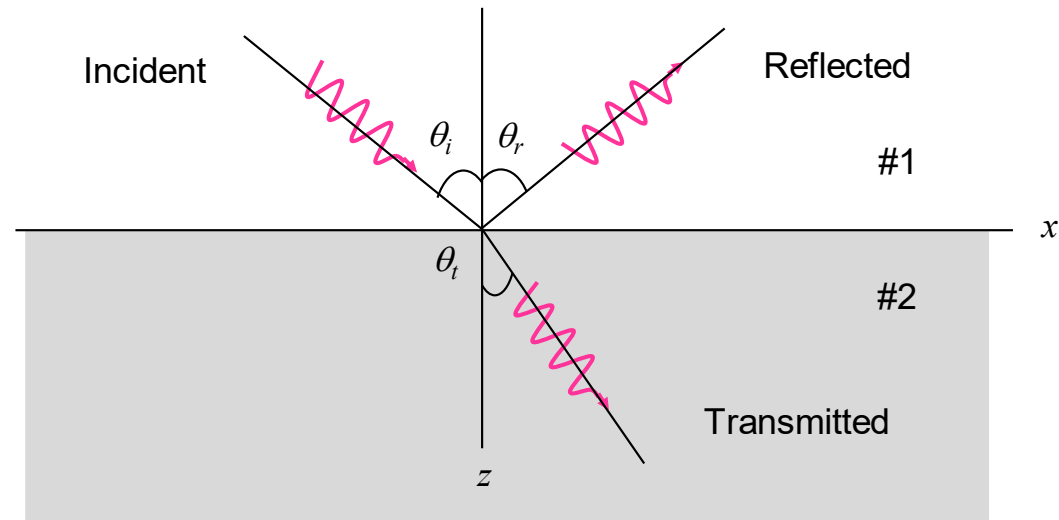
$$\underline{E}^r(x, 0) = \underline{E}_{0r} e^{-jk_{xr}x}$$

$$\underline{E}^t(x, 0) = \underline{E}_{0t} e^{-jk_{xt}x}$$

Law of Reflection

$$k_{xi} = k_{xr}$$

(This comes from the first part of the phase matching condition.)



Incident and reflected waves:

$$\begin{aligned} k_{xi} &= k_1 \sin \theta_i \\ k_{yi} &= 0 \\ k_{zi} &= k_1 \cos \theta_i \end{aligned}$$

$$\begin{aligned} k_{xr} &= k_1 \sin \theta_r \\ k_{yr} &= 0 \\ k_{zr} &= k_1 \cos \theta_r \end{aligned}$$

$$k_1 \sin \theta_i = k_1 \sin \theta_r$$



$$\theta_i = \theta_r$$

Law of reflection

Snell's Law

$$k_{xi} = k_{xt}$$

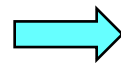
(This comes from the second part of the phase matching condition.)

Incident and transmitted waves:

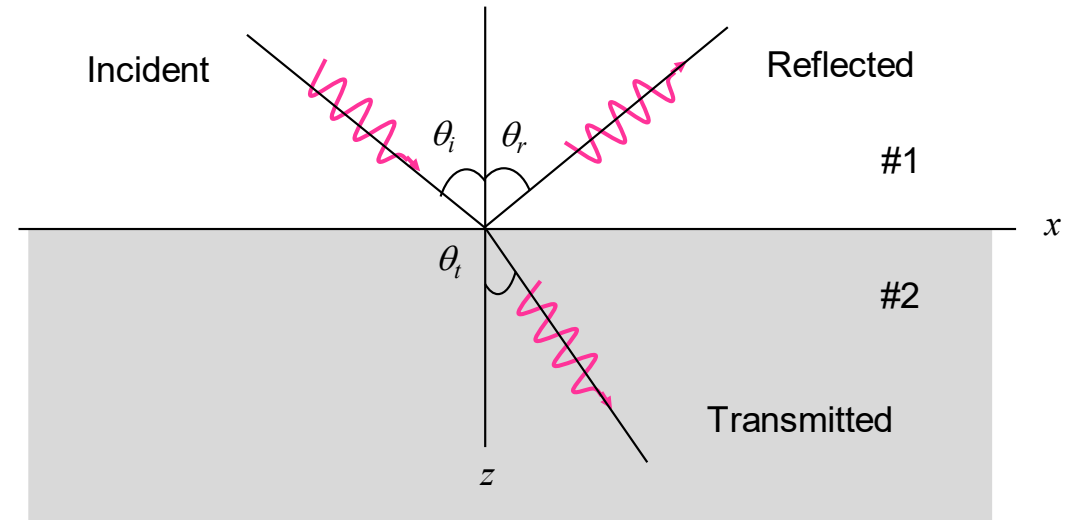
$$\begin{aligned} k_{xi} &= k_1 \sin \theta_i \\ k_{yi} &= 0 \\ k_{zi} &= k_1 \cos \theta_i \end{aligned}$$

$$\begin{aligned} k_{xt} &= k_2 \sin \theta_t \\ k_{yt} &= 0 \\ k_{zt} &= k_2 \cos \theta_t \end{aligned}$$

$$k_1 \sin \theta_i = k_2 \sin \theta_t$$



$$n_1 \sin \theta_i = n_2 \sin \theta_t \quad \text{Snell's law}$$



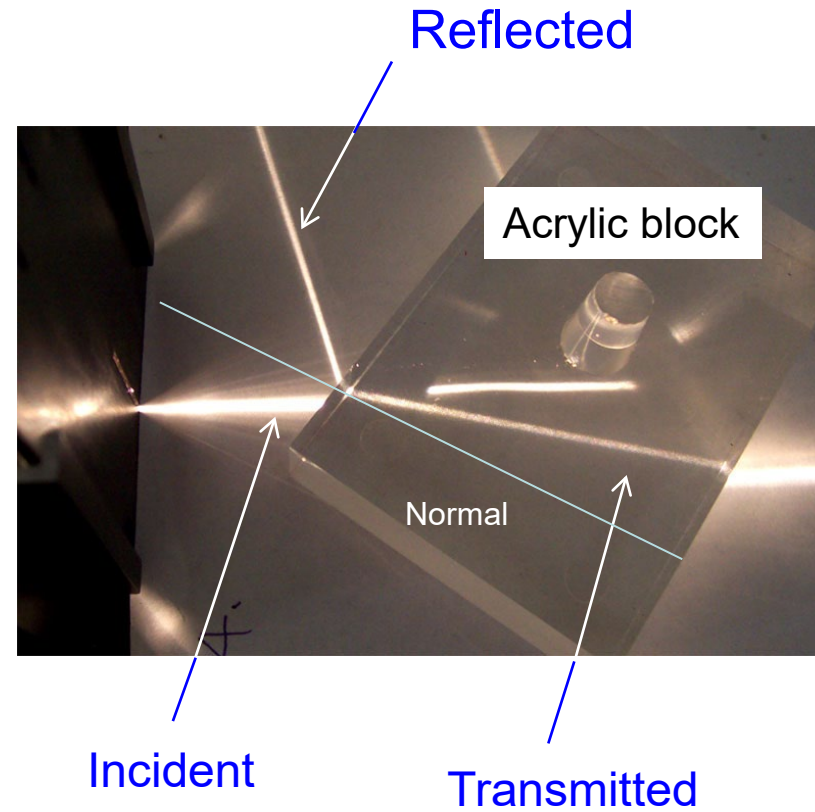
We define the **index of refraction**:

$$n_i \equiv k_i / k_0 = \omega \sqrt{\mu_i \epsilon_i} / \omega \sqrt{\mu_0 \epsilon_0} = \sqrt{\epsilon_{ri} \mu_{ri}} \\ i = 1, 2$$

Note: The wave is bent towards the normal (z axis) when entering a more optically "dense" region (a higher index of refraction).

Snell's Law (cont.)

The bending of light (or EM waves in general) is called **refraction**.



<http://en.wikipedia.org/wiki/Refraction>

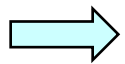
Example

Given: $\theta_i = 45^\circ$

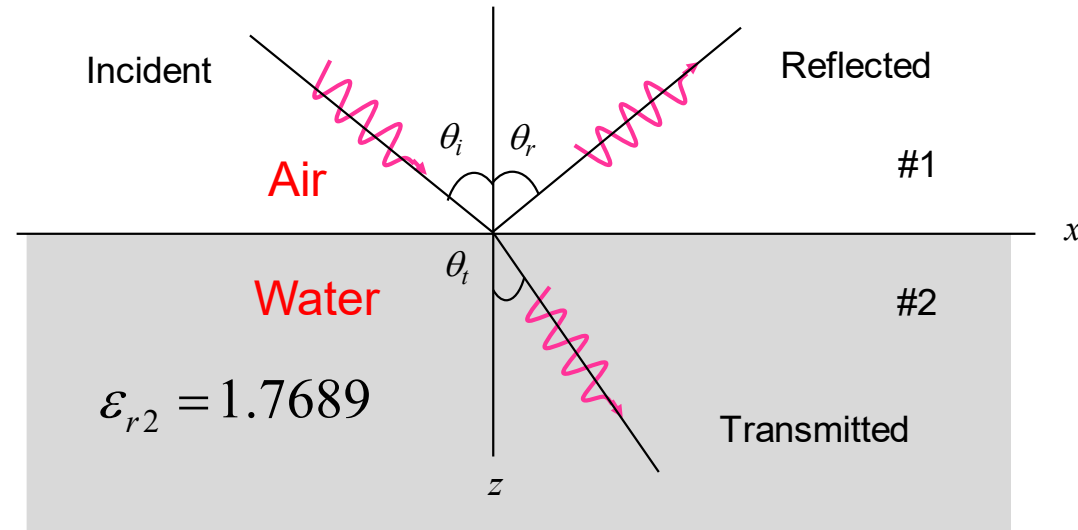
Find the transmitted angle.

$$n_1 \sin \theta_i = n_2 \sin \theta_t$$

$$\sqrt{1} \sin \theta_i = \sqrt{1.7689} \sin \theta_t$$



$$\theta_t = 32.1^\circ$$



Note that in going from a less dense to a more dense medium, the wavevector is bent towards the normal.

Note:

If the wave is incident from the water region at an incident angle of 32.1° , the wave will exit into the air region at an angle of 45° .

Note:

At microwave frequencies and below, the relative permittivity of pure water is about 81.
At optical frequencies it is about 1.7689.

Critical Angle

The wave is **incident from a more dense** region onto a less dense region.

$$n_1 > n_2$$

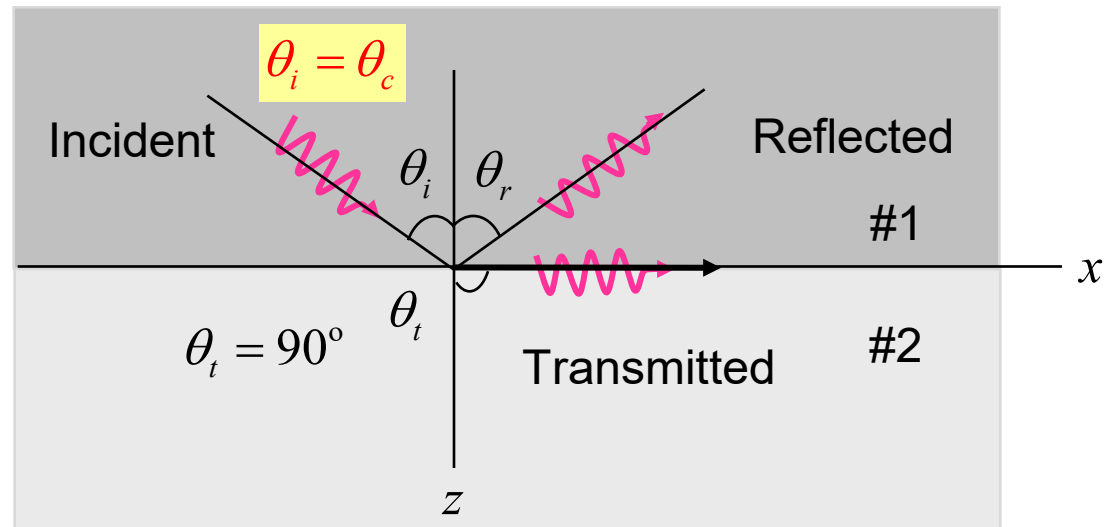
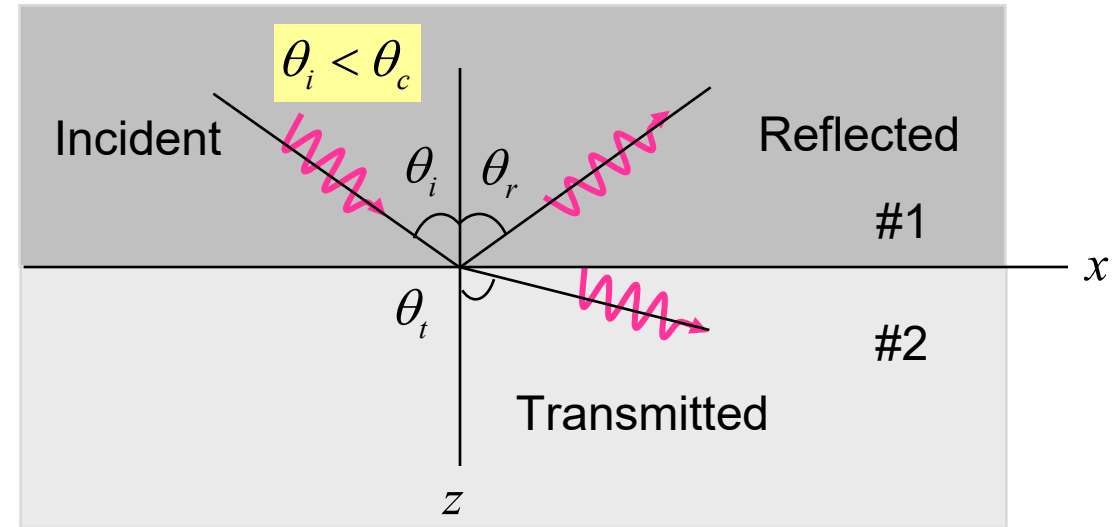
$$n_1 \sin \theta_i = n_2 \sin \theta_t$$

$$\Rightarrow \sin \theta_t = \frac{n_1}{n_2} \sin \theta_i$$

At the critical angle: $\theta_t = 90^\circ$

$$\sin(90^\circ) = \frac{n_1}{n_2} \sin \theta_c$$

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$



$$\theta_i = \theta_c : k_{xi} = k_{xr} = k_{xt} = k_2 \sin \theta_t = k_2$$

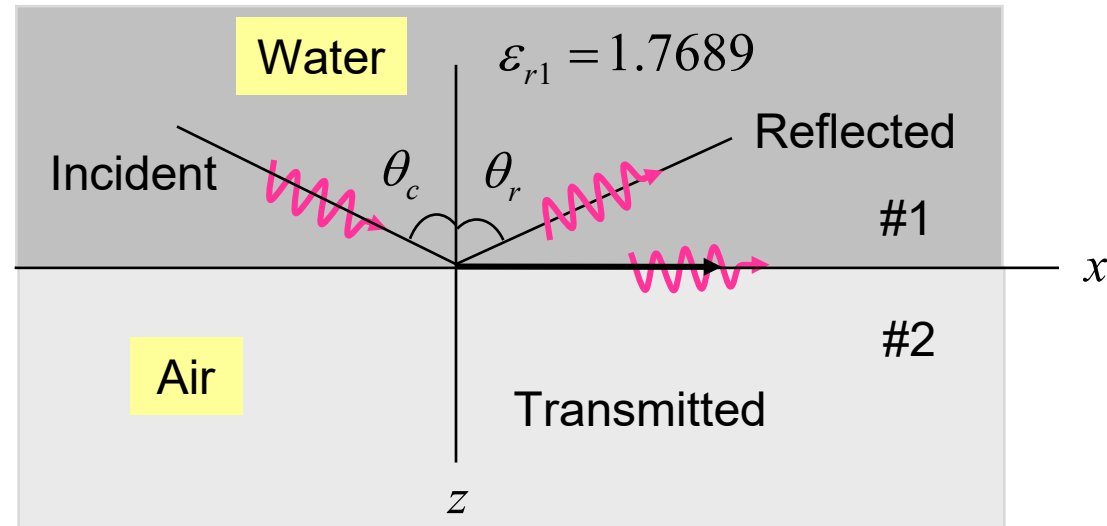
Critical Angle (cont.)

Example

Find the critical angle.

$$\begin{aligned}\theta_c &= \sin^{-1}\left(\frac{n_2}{n_1}\right) \\ &= \sin^{-1}\left(\frac{\sqrt{1}}{\sqrt{1.7689}}\right)\end{aligned}$$

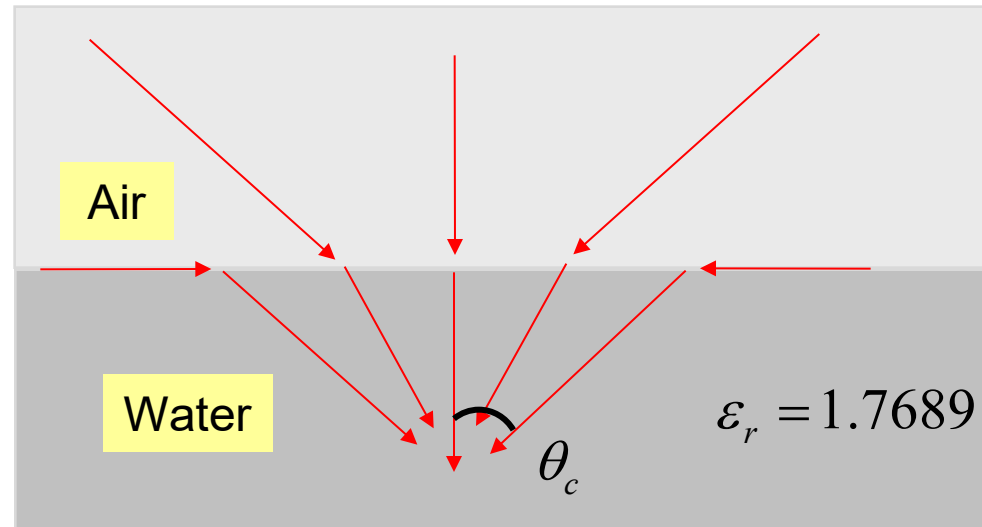
$$\theta_c = 48.75^\circ$$



Critical Angle (cont.)

Example: "fish-eye" effect

$$\theta_c = 48.75^\circ$$



The critical angle explains the “fish-eye” effect that you can observe in a swimming pool.

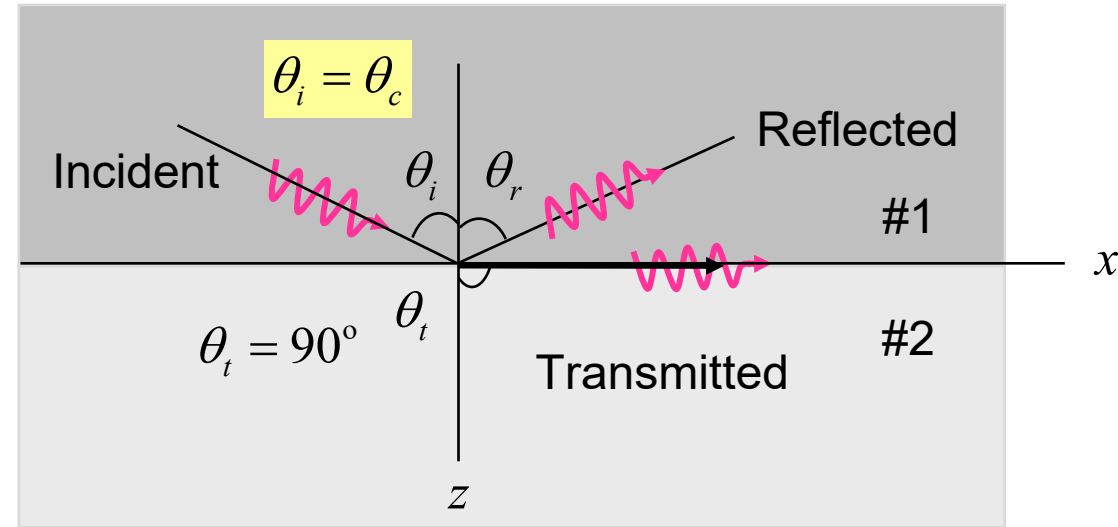
A fish can see everything above the water by only looking no farther than 49° from the vertical!

Critical Angle (cont.)

At the critical angle:

$$k_{xi} = k_{xr} = k_{xt} = k_2 \sin \theta_t = k_2$$

$$\begin{aligned} k_{zt} &= \sqrt{k_2^2 - k_{xt}^2} \\ &= \sqrt{k_2^2 - k_2^2} \\ &= 0 \end{aligned}$$



- ❖ There is no vertical variation of the field in the less-dense (transmitted) region.
- ❖ There are still fields in the less dense region, however.

Critical Angle (cont.)

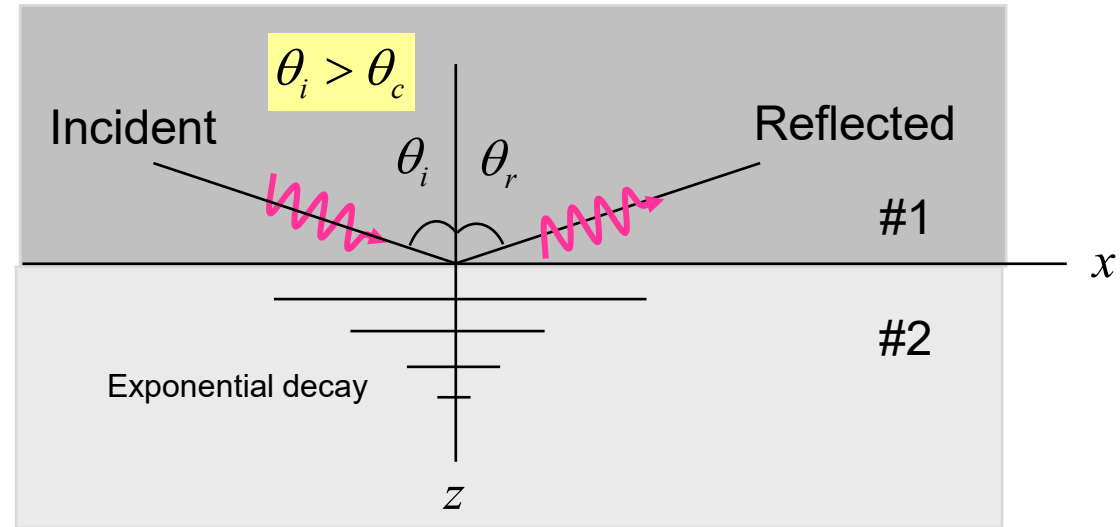
Beyond the critical angle:

$$\theta_i > \theta_c$$

$$\Rightarrow k_{xi} > k_2$$

$$(k_{xi} = k_1 \sin \theta_i > k_1 \sin \theta_c = k_2 \sin 90^\circ = k_2)$$

$$\begin{aligned} k_{zt} &= (k_2^2 - k_{xt}^2)^{1/2} \\ &= (k_2^2 - k_{xi}^2)^{1/2} \\ &= -j\sqrt{k_{xi}^2 - k_2^2} \\ &= -j\sqrt{k_1^2 \sin^2 \theta_i - k_2^2} \\ &= -jk_0\sqrt{n_1^2 \sin^2 \theta_i - n_2^2} \\ &= -j\alpha_{zt} \end{aligned}$$



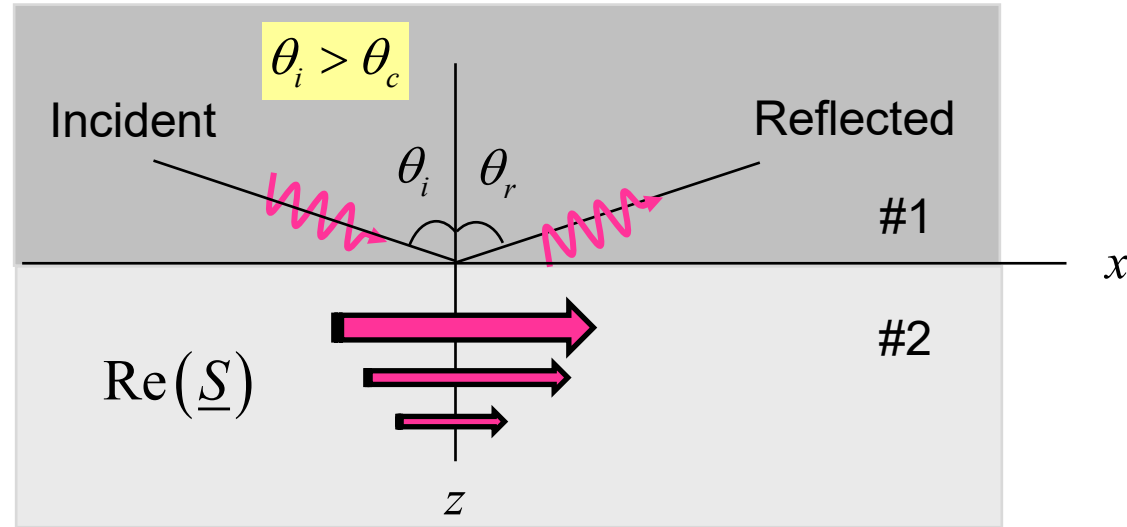
There is an **exponential decay** of the field in the vertical direction in the less-dense region.

$$\psi_t(x, z) \equiv e^{-j(k_{xt}x + k_{zt}z)} = e^{-j(k_{xt}x)} e^{-\alpha_{zt}z}$$

$$\alpha_{zt} = k_0 \sqrt{n_1^2 \sin^2 \theta_i - n_2^2}$$

Critical Angle (cont.)

Beyond the critical angle:



The power flows completely horizontally in the lower region.

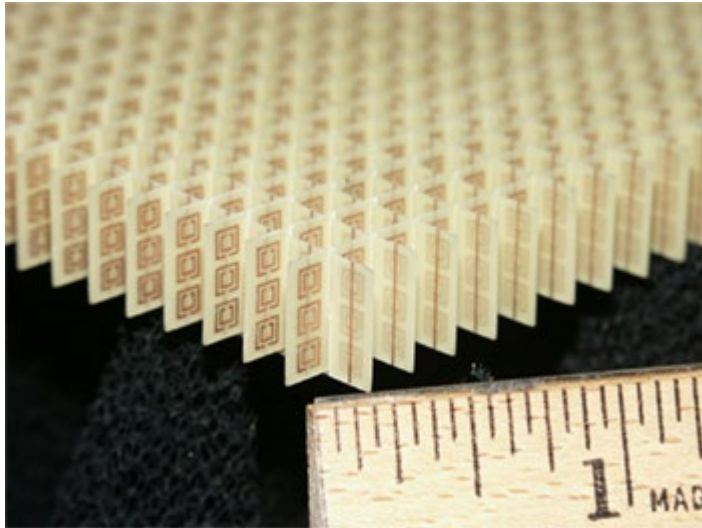
No power crosses the boundary and enters into the less dense region.

This must be true from conservation of energy, since the field decays exponentially in the lossless region 2.

Exotic Materials

Artificial “metamaterials” that have been designed that have exotic permittivity and/or permeability performance.

<https://en.wikipedia.org/wiki/Metamaterial>

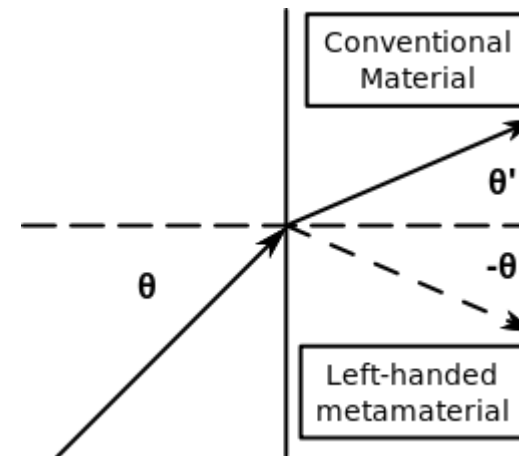


Negative index metamaterial array configuration, which was constructed of copper split-ring resonators and wires mounted on interlocking sheets of fiberglass circuit board. The total array consists of 3 by 20×20 unit cells with overall dimensions of 10×100×100 mm.

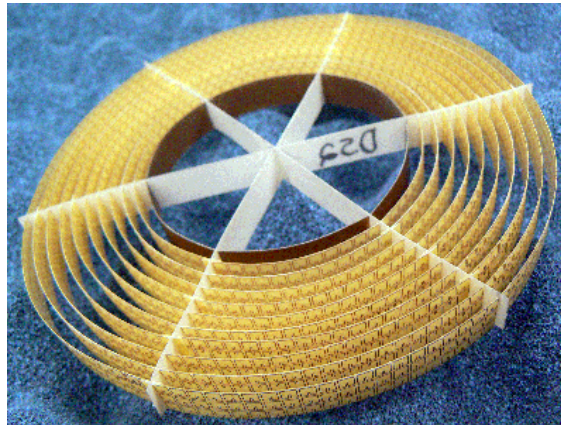
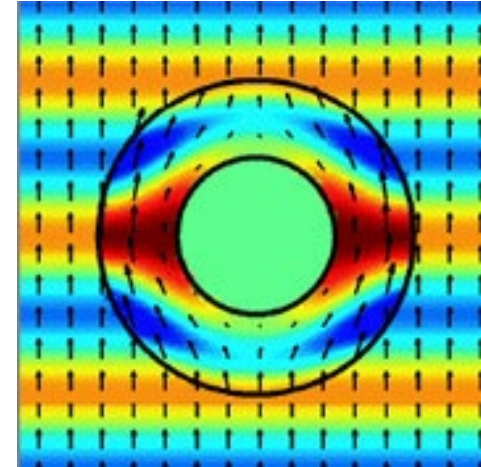
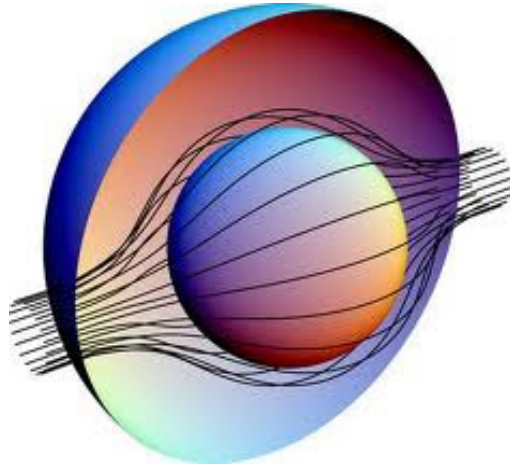
$$\varepsilon_r < 0$$

$$\mu_r < 0$$

(over a certain bandwidth of operation)



Exotic Materials (cont.)

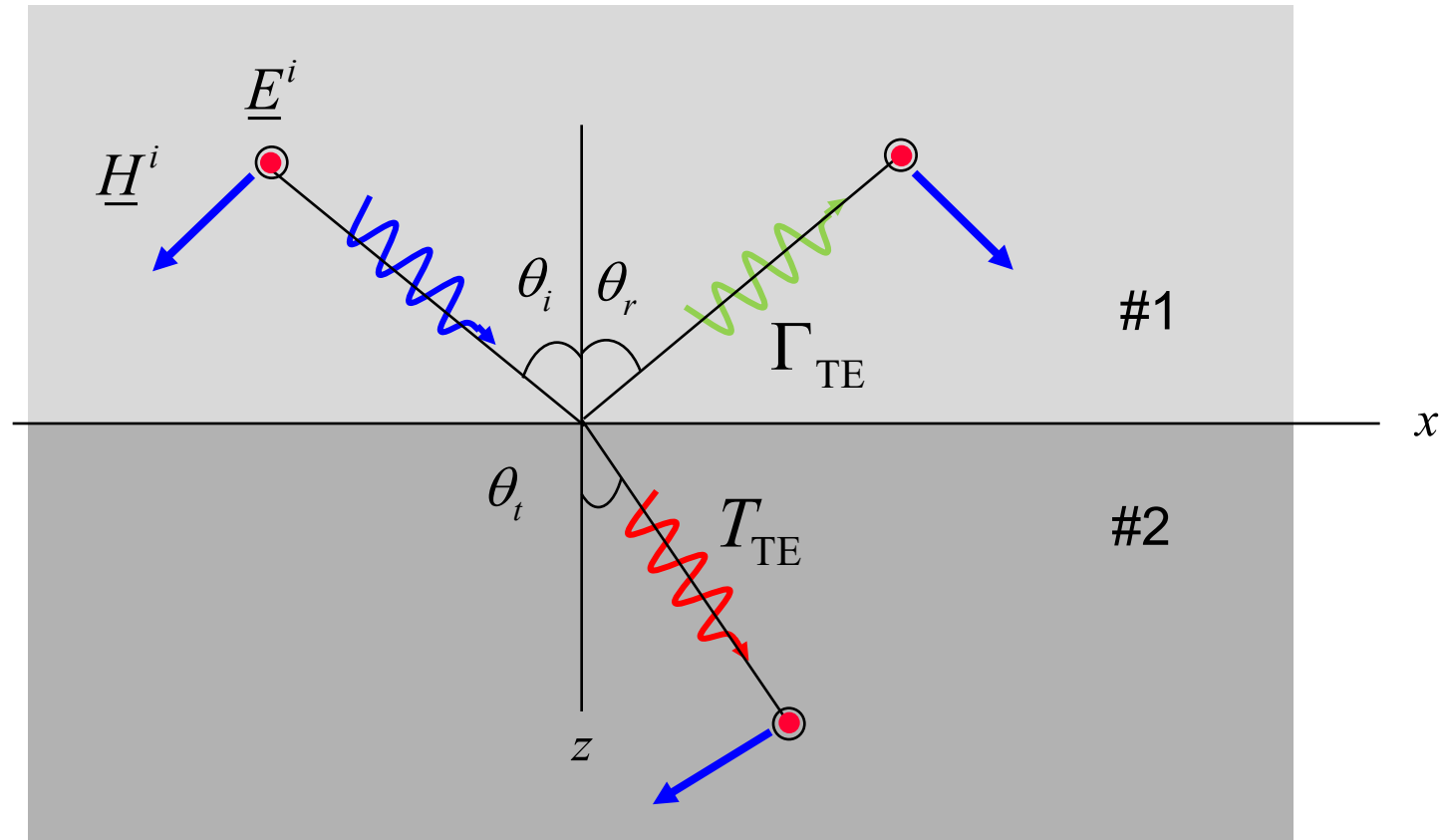


The Duke cloaking device masks an object at one microwave frequency.

Image courtesy Dr. David R. Smith.

Cloaking of objects is one area of research in metamaterials.

TE_z Reflection



Note that the electric field vector is in the y direction.

TE_z $\Leftrightarrow$ polarized perpendicular to the plane of incidence

TE_z Reflection (cont.)

Incident Wave

$$\underline{\underline{E}}^i = \hat{y} E_0 e^{-jk_{xi}x - jk_{zi}z}$$

Reflected Wave

$$\underline{\underline{E}}^r = \hat{y} \Gamma_{\text{TE}} E_0 e^{-jk_{xr}x + jk_{zr}z}$$

Transmitted Wave

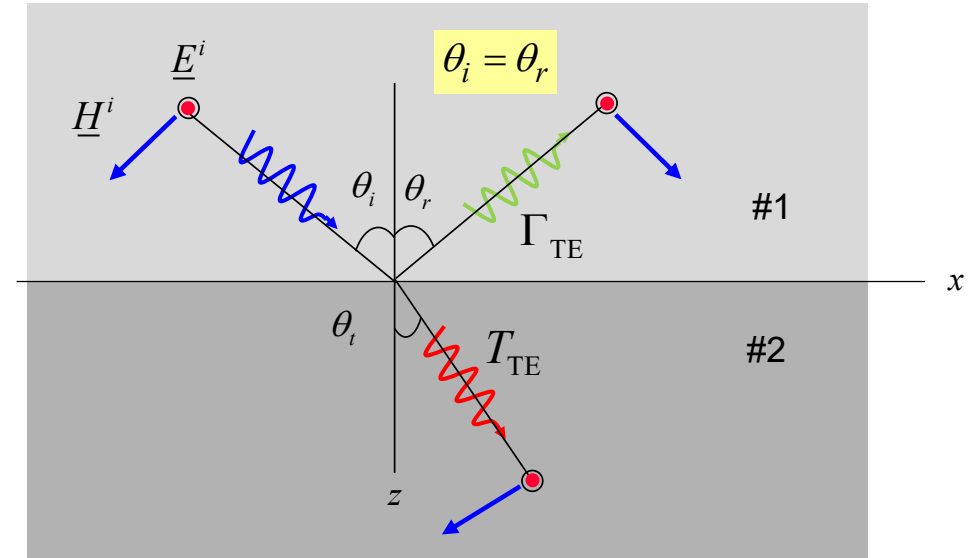
$$\underline{\underline{E}}^t = \hat{y} T_{\text{TE}} E_0 e^{-jk_{xt}x - jk_{zt}z}$$

where

Γ_{TE} = Reflection Coefficient

T_{TE} = Transmission Coefficient

$$k_{xi} = k_{xr} = k_{xt} = k_1 \sin \theta_i$$



Incident Wave

$$k_{zi} = k_1 \cos \theta_i$$

Reflected Wave

$$k_{zr} = k_1 \cos \theta_r$$

Transmitted Wave

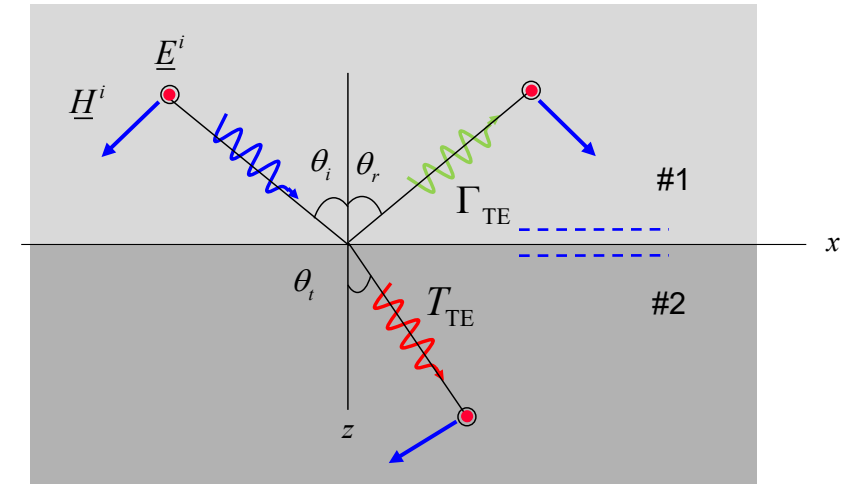
$$k_{zt} = k_2 \cos \theta_t$$

TE_z Reflection (cont.)

Recall that the tangential component of the electric field must be continuous at an interface.

Boundary condition at $z = 0$:

$$E_y^i + E_y^r = E_y^t$$



$$\text{At } z = 0: \quad E_0 e^{-jk_{xi}x - jk_{zi}z} + \Gamma_{TE} E_0 e^{-jk_{xr}x + jk_{zr}z} = T_{TE} E_0 e^{-jk_{xt}x - jk_{zt}z}$$

$$\downarrow z = 0$$

$$E_0 e^{-jk_{xi}x} + \Gamma_{TE} E_0 e^{-jk_{xr}x} = T_{TE} E_0 e^{-jk_{xt}x}$$

$$\downarrow k_{xi} = k_{xr} = k_{xt}$$

$$E_0 + \Gamma_{TE} E_0 = T_{TE} E_0$$

$$\downarrow \text{Divide by } E_0$$

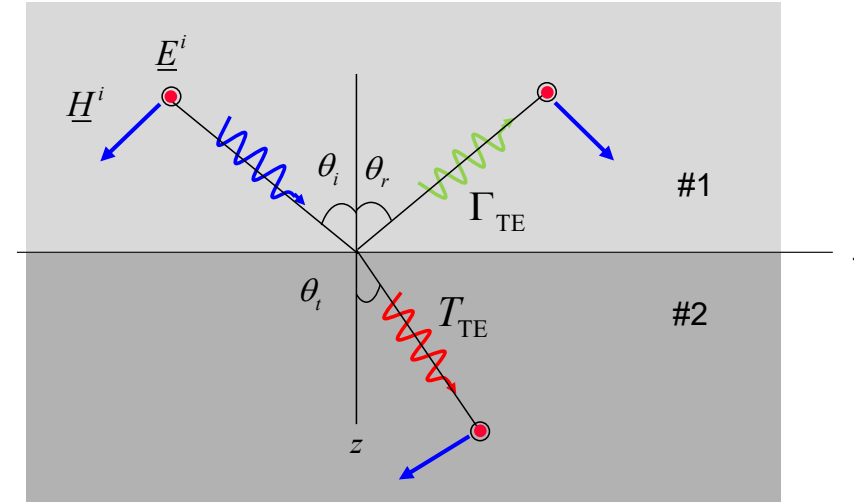
$$1 + \Gamma_{TE} = T_{TE}$$

TE_z Reflection (cont.)

We now look at the magnetic fields.

$$\nabla \times \underline{E} = -j\omega\mu\underline{H}$$

$$\Rightarrow \underline{H} = \left(\frac{-1}{j\omega\mu} \right) \nabla \times \underline{E}$$



$$\underline{E}^i = \hat{y} E_0 e^{-j(k_{xi}x + k_{zi}z)}$$

$$\underline{H}^i = (-\hat{x} k_{zi} + \hat{z} k_{xi}) \left(\frac{E_0}{\omega\mu_1} \right) e^{-j(k_{xi}x + k_{zi}z)}$$

$$\underline{E}^r = \hat{y} \Gamma_{TE} E_0 e^{-j(k_{xr}x - k_{zr}z)}$$

$$\underline{H}^r = (+\hat{x} k_{zr} + \hat{z} k_{xr}) \left(\frac{\Gamma_{TE} E_0}{\omega\mu_1} \right) e^{-j(k_{xr}x - k_{zr}z)}$$

$$\underline{E}^t = \hat{y} T_{TE} E_0 e^{-j(k_{xt}x + k_{zt}z)}$$

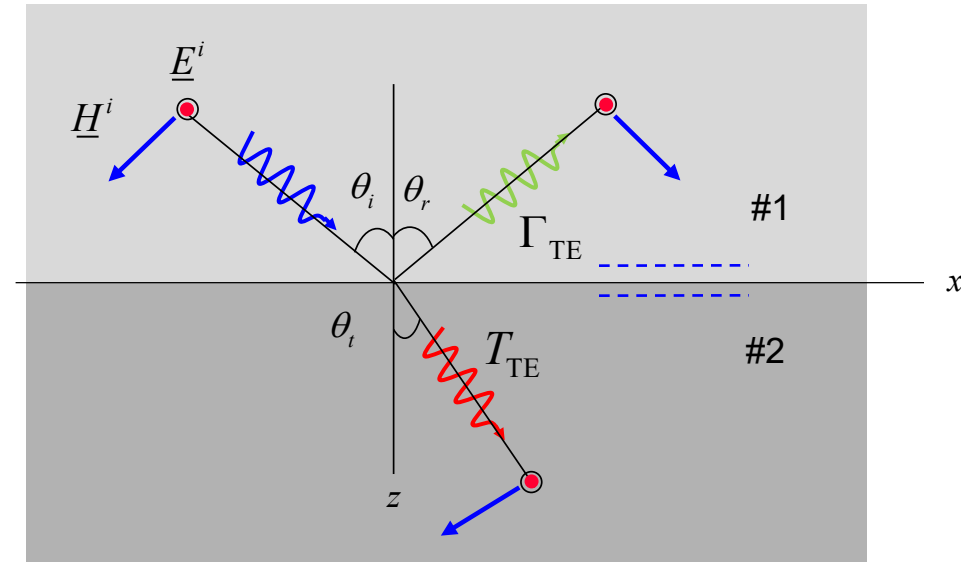
$$\underline{H}^t = (-\hat{x} k_{zt} + \hat{z} k_{xt}) \frac{T_{TE} E_0}{\omega\mu_2} e^{-j(k_{xt}x + k_{zt}z)}$$

TE_z Reflection (cont.)

Recall that the tangential component of the magnetic field must be continuous at an interface (no surface currents).

Boundary condition at $z = 0$:

$$H_x^i + H_x^r = H_x^t$$



Hence, we have:

$$-\frac{k_{zi}}{\omega\mu_1} + \frac{k_{zr}\Gamma_{TE}}{\omega\mu_1} = -\frac{k_{zt}T_{TE}}{\omega\mu_2}$$

$$H_x^i = (-k_{zi}) \left(\frac{E_0}{\omega\mu_1} \right) e^{-j(k_{xi}x + k_{zi}z)}$$

$$H_x^r = (k_{zr}) \left(\frac{\Gamma_{TE}E_0}{\omega\mu_1} \right) e^{-j(k_{xr}x - k_{zr}z)}$$

$$H_x^t = (-k_{zt}) \frac{T_{TE}E_0}{\omega\mu_2} e^{-j(k_{xt}x + k_{zt}z)}$$

TE_z Reflection (cont.)

Enforcing both boundary conditions, we have:

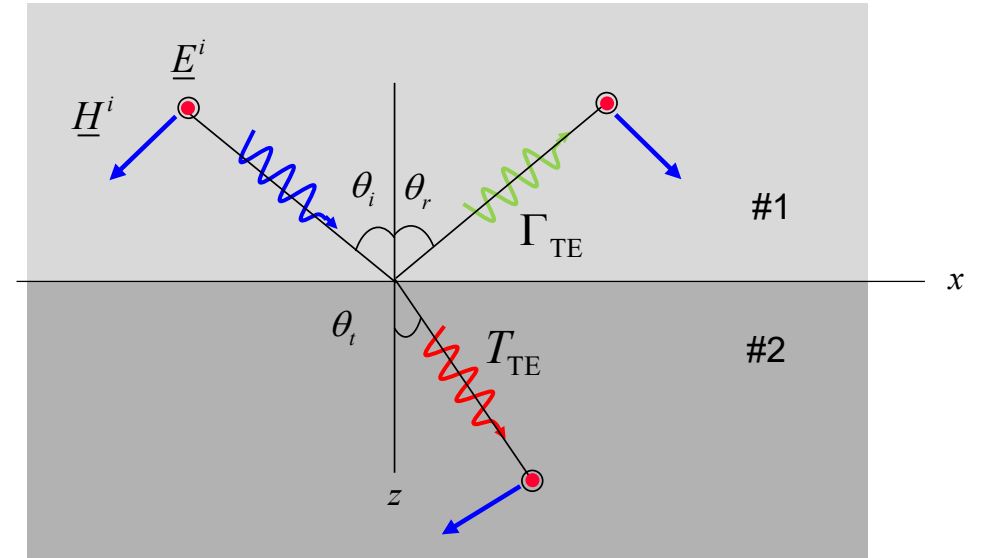
$$1 + \Gamma_{\text{TE}} = T_{\text{TE}}$$

$$\frac{k_{zi}}{\omega\mu_1} - \frac{k_{zr}\Gamma_{\text{TE}}}{\omega\mu_1} = \frac{k_{zt}T_{\text{TE}}}{\omega\mu_2}$$

The solution is:

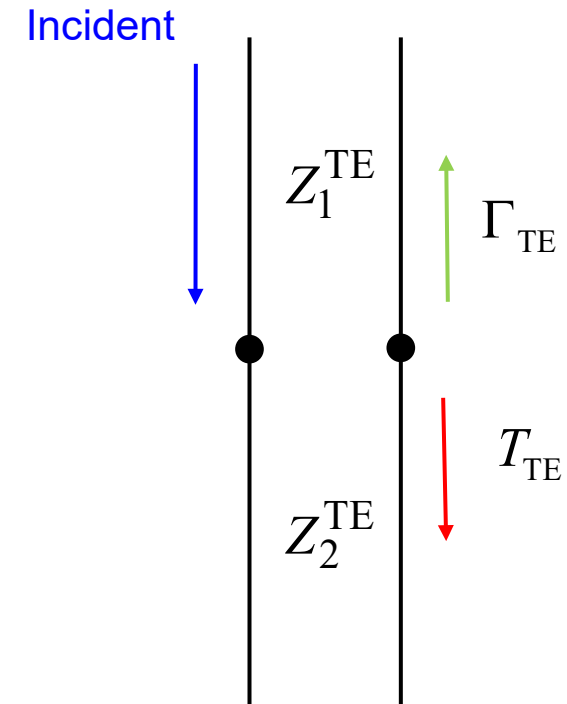
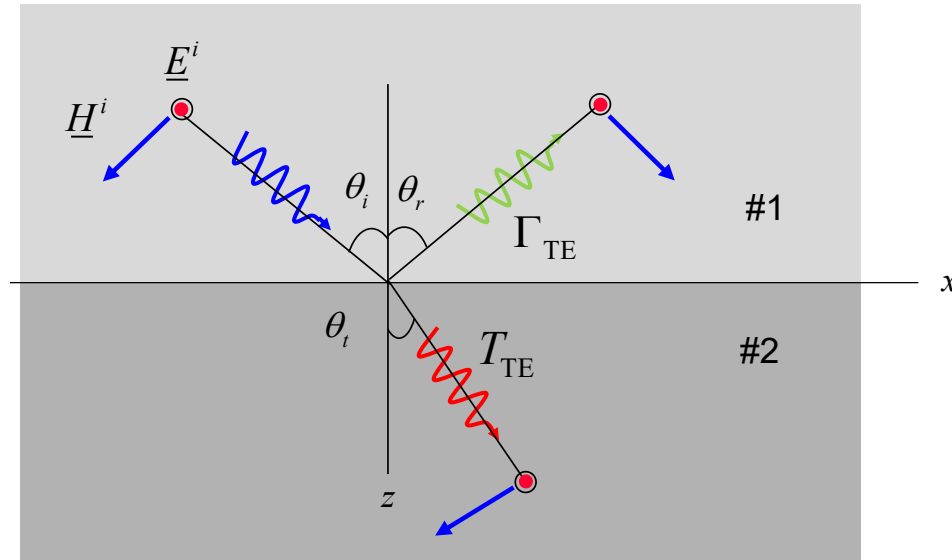
$$\Gamma_{\text{TE}} = \frac{\left(\frac{\omega\mu_2}{k_{zt}}\right) - \left(\frac{\omega\mu_1}{k_{zi}}\right)}{\left(\frac{\omega\mu_2}{k_{zt}}\right) + \left(\frac{\omega\mu_1}{k_{zi}}\right)}$$

$$T_{\text{TE}} = 1 + \Gamma_{\text{TE}}$$



TE_z Reflection (cont.)

Transmission Line Model

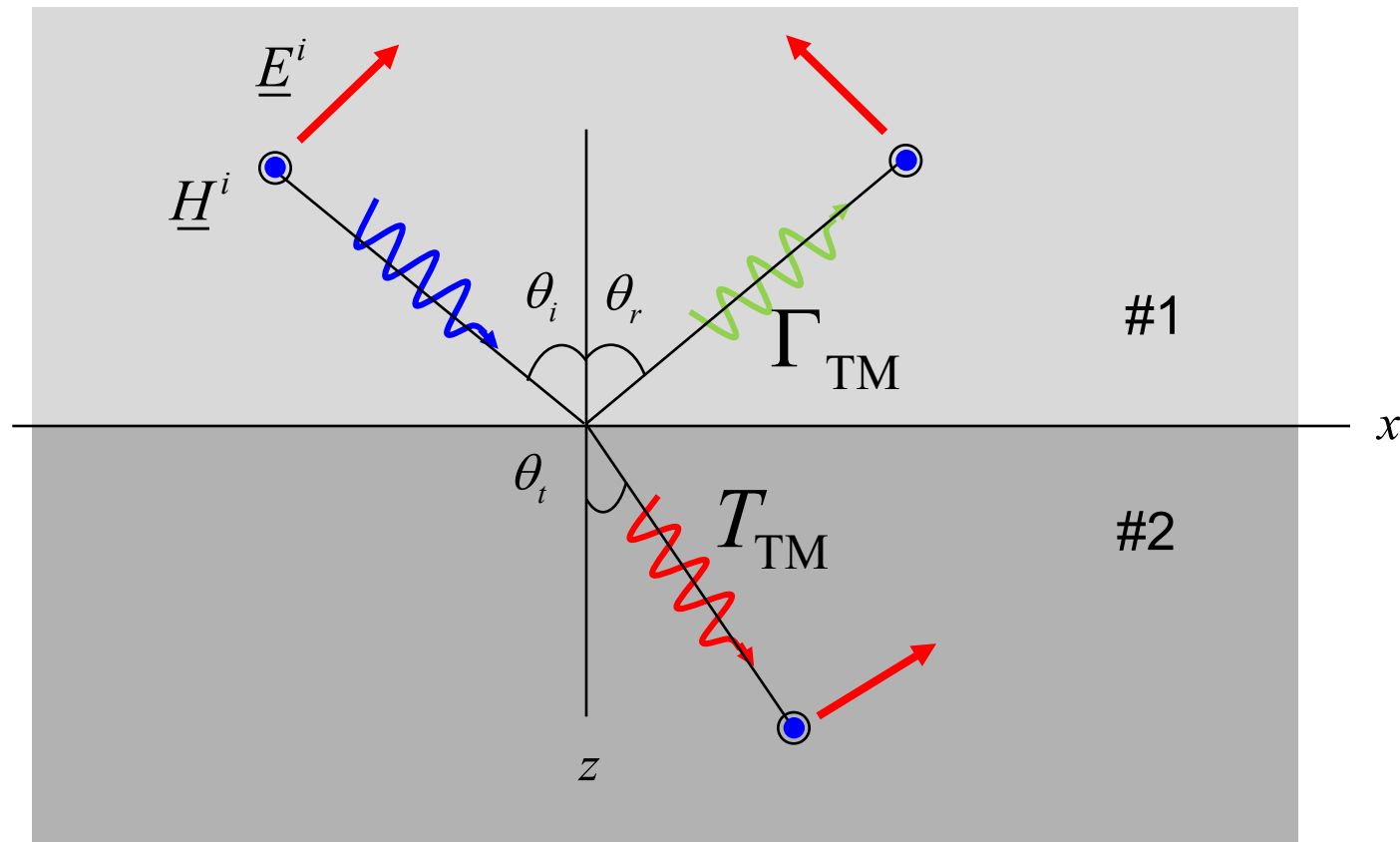


$$\Gamma_{TE} = \frac{Z_2^{TE} - Z_1^{TE}}{Z_2^{TE} + Z_1^{TE}}$$

$$T_{TE} = 1 + \Gamma_{TE}$$

$$Z_1^{TE} = \left(\frac{\omega \mu_1}{k_{zi}} \right) \quad Z_2^{TE} = \left(\frac{\omega \mu_2}{k_{zt}} \right)$$

TM_z Reflection



Note that the electric field vector is in the xz plane.

$\text{TE}_z \Leftrightarrow$ polarized parallel to the plane of incidence

TM_z Reflection (cont.)

Incident Wave

$$\underline{H}^i = \underline{\hat{y}} H_0 \left(\frac{\omega \epsilon_1}{k_{zi}} \right) e^{-jk_{xi}x - jk_{zi}z}$$

Reflected Wave

$$\underline{H}^r = \underline{\hat{y}} (-\Gamma_{\text{TM}}) H_0 \left(\frac{\omega \epsilon_1}{k_{zr}} \right) e^{-jk_{xr}x + jk_{zr}z}$$

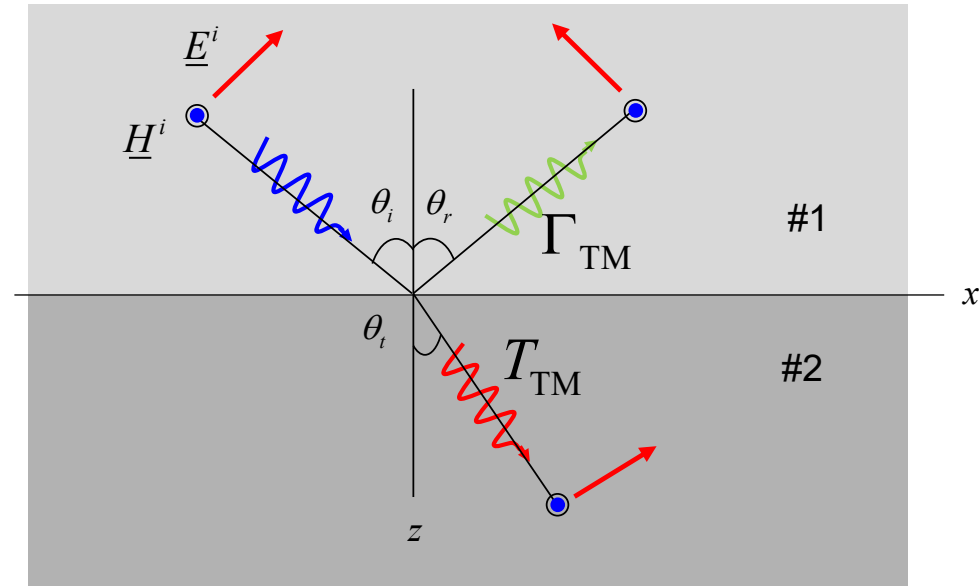
Transmitted Wave

$$\underline{H}^t = \underline{\hat{y}} T_{\text{TM}} H_0 \left(\frac{\omega \epsilon_2}{k_{zt}} \right) e^{-jk_{xt}x - jk_{zt}z}$$

where

Γ_{TE} = Reflection Coefficient

T_{TE} = Transmission Coefficient



Incident Wave

$$k_{zi} = k_1 \cos \theta_i$$

Reflected Wave

$$k_{zr} = k_1 \cos \theta_r$$

Transmitted Wave

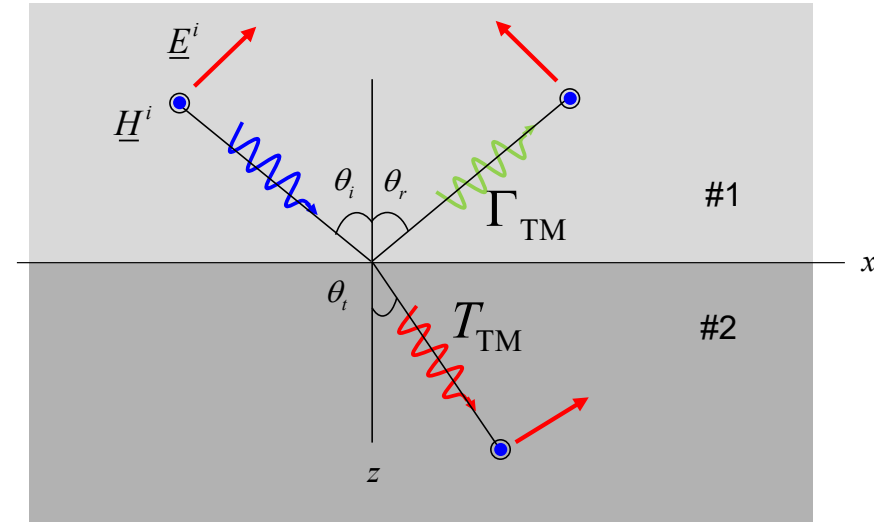
$$k_{zt} = k_2 \cos \theta_t$$

TM_z Reflection (cont.)

We now look at the electric fields.

$$\nabla \times \underline{H} = j\omega\epsilon \underline{E}$$

$$\Rightarrow \underline{E} = \left(\frac{1}{j\omega\epsilon} \right) \nabla \times \underline{H}$$



$$\underline{H}^i = \underline{\hat{y}} H_0 \left(\frac{\omega\epsilon_1}{k_{zi}} \right) e^{-jk_{xi}x - jk_{zi}z}$$

$$\underline{E}^i = (\underline{\hat{x}} - \underline{\hat{z}} k_{xi} / k_{zi}) H_0 e^{-j(k_{xi}x + k_{zi}z)}$$

$$\underline{H}^r = \underline{\hat{y}} (-\Gamma_{\text{TM}}) \left(\frac{\omega\epsilon_1}{k_{zr}} \right) H_0 e^{-jk_{xr}x + jk_{zr}z}$$

$$\underline{E}^r = (\underline{\hat{x}} + \underline{\hat{z}} k_{xr} / k_{zr}) \Gamma_{\text{TM}} H_0 e^{-j(k_{xr}x - k_{zr}z)}$$

$$\underline{H}^t = \underline{\hat{y}} T_{\text{TM}} \left(\frac{\omega\epsilon_2}{k_{zt}} \right) H_0 e^{-jk_{xt}x - jk_{zt}z}$$

$$\underline{E}^t = (\underline{\hat{x}} - \underline{\hat{z}} k_{xt} / k_{zt}) T_{\text{TM}} H_0 e^{-j(k_{xt}x + k_{zt}z)}$$

Note that Γ_{TM} is the reflection coefficient for the tangential electric field (E_x).

TM_z Reflection (cont.)

Boundary conditions:

$$H_y^i + H_y^r = H_y^t$$

$$E_x^i + E_x^r = E_x^t$$

Enforcing both boundary conditions, we have

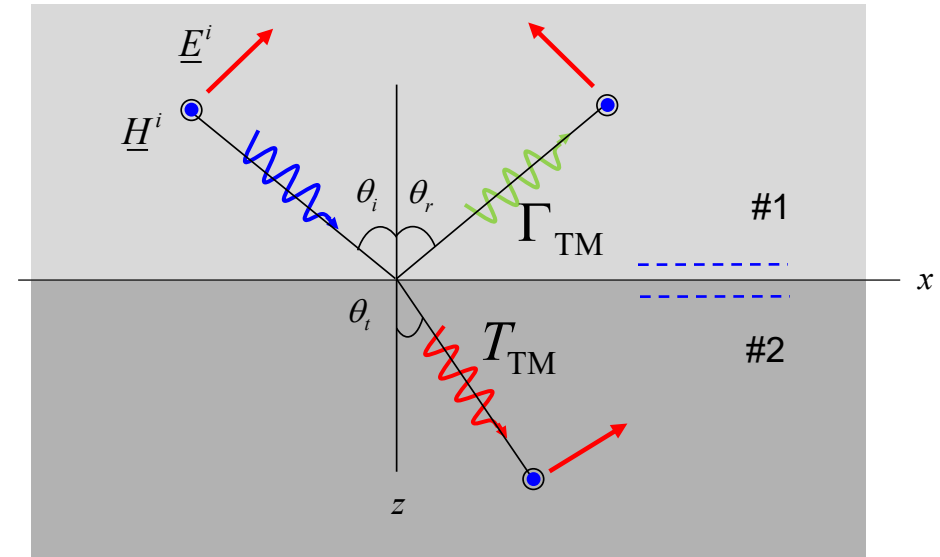
$$\frac{\omega \epsilon_1}{k_{zi}} - \Gamma_{\text{TM}} \left(\frac{\omega \epsilon_1}{k_{zr}} \right) = T_{\text{TM}} \left(\frac{\omega \epsilon_2}{k_{zt}} \right)$$

$$1 + \Gamma_{\text{TM}} = T_{\text{TM}}$$

The solution is:

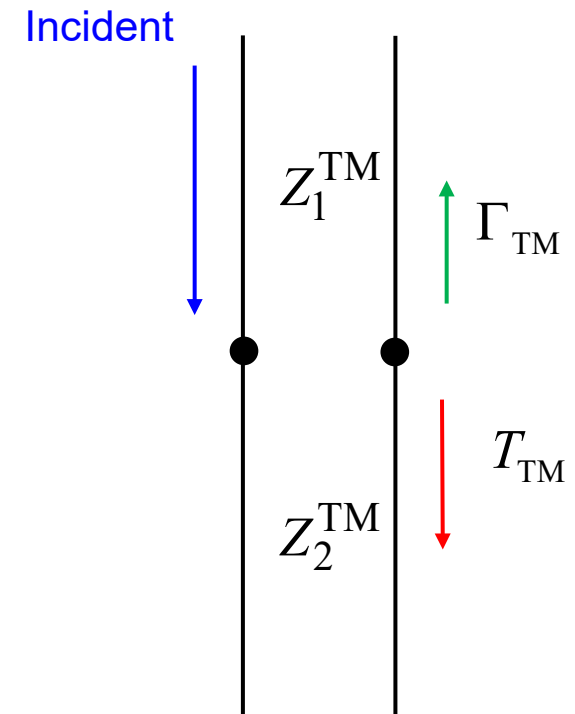
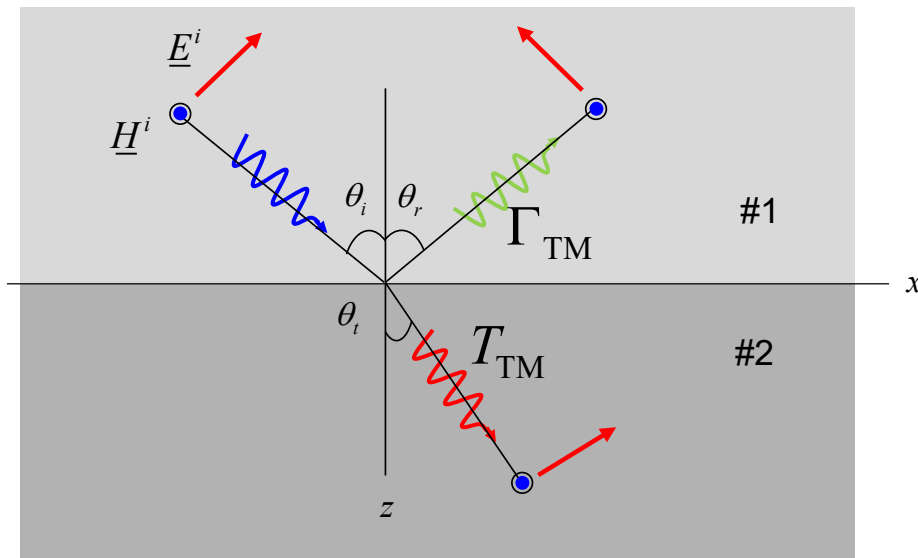
$$\Gamma_{\text{TM}} = \frac{\left(\frac{k_{zt}}{\omega \epsilon_2} \right) - \left(\frac{k_{zi}}{\omega \epsilon_1} \right)}{\left(\frac{k_{zt}}{\omega \epsilon_2} \right) + \left(\frac{k_{zi}}{\omega \epsilon_1} \right)}$$

$$T_{\text{TM}} = 1 + \Gamma_{\text{TM}}$$



TM_z Reflection (cont.)

Transmission Line Model



$$\Gamma_{TM} = \frac{Z_2^{TM} - Z_1^{TM}}{Z_2^{TM} + Z_1^{TM}}$$

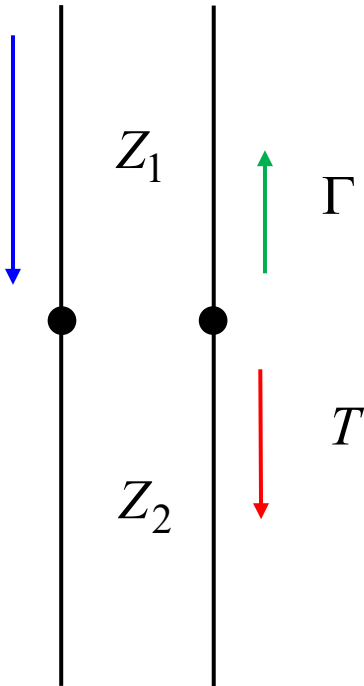
$$T_{TM} = 1 + \Gamma_{TM}$$

$$Z_1^{TM} = \left(\frac{k_{zi}}{\omega \epsilon_1} \right) \quad Z_2^{TM} = \left(\frac{k_{zt}}{\omega \epsilon_2} \right)$$

TM_z Reflection (cont.)

Summary of Transmission Line Modeling Equations

Incident

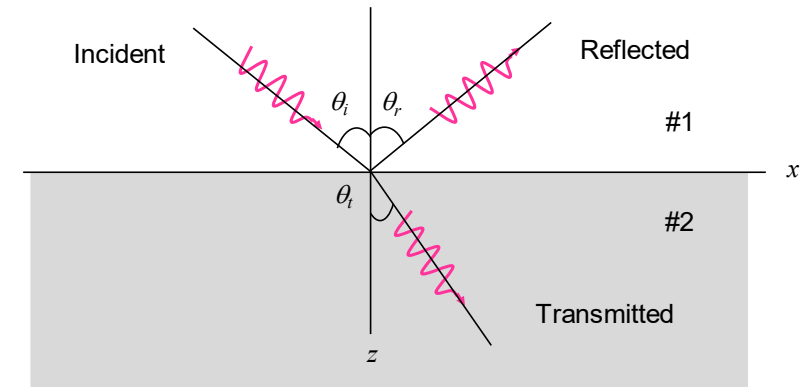


$$Z_1^{\text{TM}} = \frac{k_{z1}}{\omega \epsilon_1}$$

$$Z_2^{\text{TM}} = \frac{k_{z2}}{\omega \epsilon_2}$$

$$Z_1^{\text{TE}} = \frac{\omega \mu_1}{k_{z1}}$$

$$Z_2^{\text{TE}} = \frac{\omega \mu_2}{k_{z2}}$$



$$k_{z1} = k_{zi} = \sqrt{k_1^2 - k_{xi}^2} = \sqrt{k_1^2 - (k_1 \sin \theta_i)^2} = k_1 \cos \theta_i$$

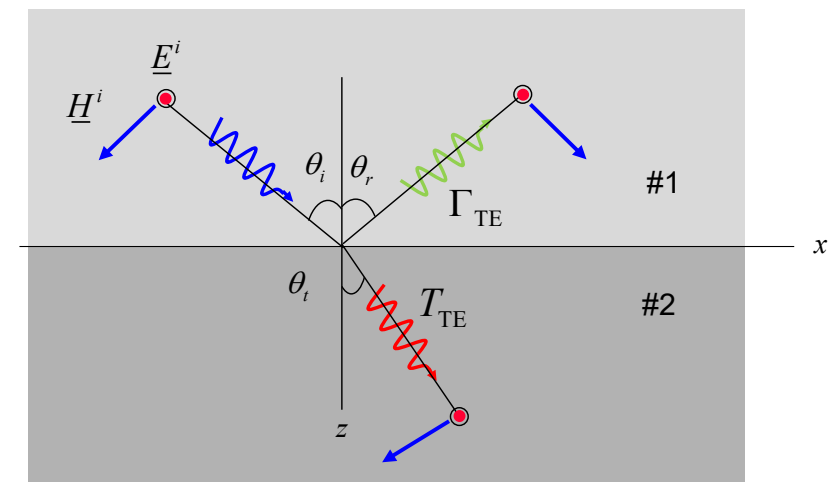
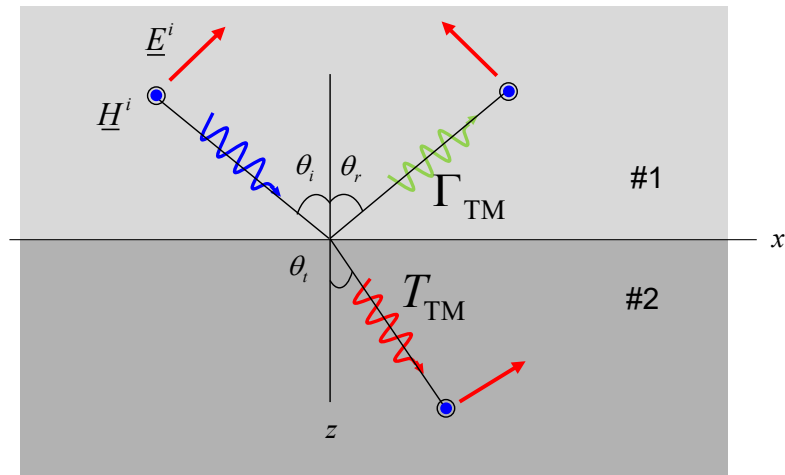
$$k_{z2} = k_{zt} = \sqrt{k_2^2 - k_{xt}^2} = \sqrt{k_2^2 - k_{xi}^2} = \sqrt{k_2^2 - (k_1 \sin \theta_i)^2} = \sqrt{k_2^2 - (k_2 \sin \theta_t)^2} = k_2 \cos \theta_t$$

Recall:

$$k_{xt} = k_{xi}$$

$$k_1 \sin \theta_i = k_2 \sin \theta_t$$

Power Reflection

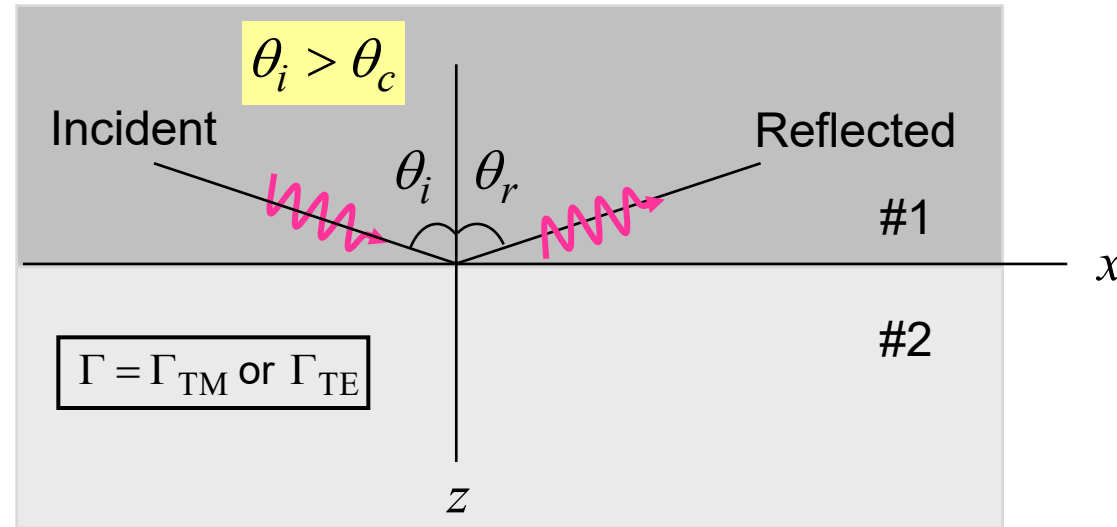


$$\% \text{ power reflected} = 100 |\Gamma|^2$$

$$\% \text{ power transmitted} = 100 (1 - |\Gamma|^2)$$

$$\Gamma = \Gamma_{TM} \text{ or } \Gamma_{TE}$$

Power Reflection Beyond Critical Angle



$$\% \text{ power reflected} = 100 |\Gamma|^2$$

$$\Gamma = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

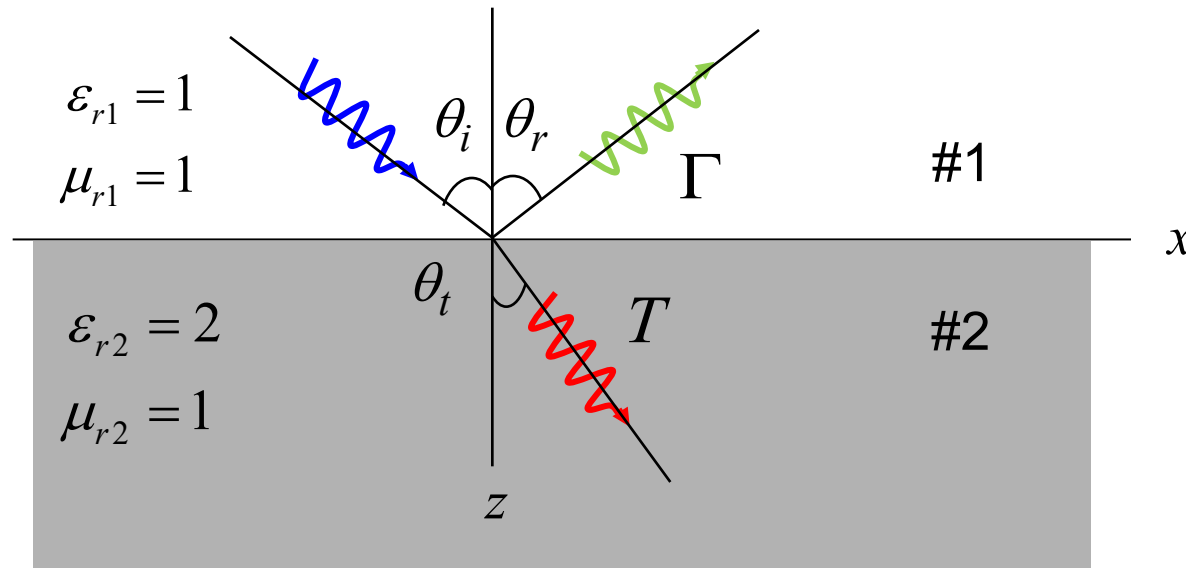
$$\left. \begin{aligned} Z_2^{\text{TM}} &= \frac{k_{zt}}{\omega \epsilon_2} \\ Z_2^{\text{TE}} &= \frac{\omega \mu_2}{k_{zt}} \end{aligned} \right\} k_{zt} = -j\alpha_{zt} \Rightarrow Z_2 = \text{imaginary ("reactive load")}$$

$$\Rightarrow |\Gamma| = 1 \quad \boxed{\text{All of the incident power is reflected.}}$$

Example

Given:

$$\theta_i = 30^\circ$$



Find:

% power reflected and transmitted for a TM_z wave

% power reflected and transmitted for a TE_z wave

Snell's law: $n_1 \sin \theta_i = n_2 \sin \theta_t$

$$\sqrt{1} \sin 30^\circ = \sqrt{2} \sin \theta_t$$

$$\theta_t = 20.70^\circ$$

Recall:

$$n_i \equiv \sqrt{\epsilon_{ri} \mu_{ri}}$$

Example (cont.)

First look at the TM_z case:

$$\Gamma_{\text{TM}} = \frac{Z_2^{\text{TM}} - Z_1^{\text{TM}}}{Z_2^{\text{TM}} + Z_1^{\text{TM}}}$$

$$\begin{aligned} Z_2^{\text{TM}} &= \frac{k_{zt}}{\omega \epsilon_2} = \frac{k_2 \cos \theta_t}{\omega \epsilon_2} = \cancel{\frac{\omega \sqrt{\mu_2 \epsilon_2} \cos \theta_t}{\omega \epsilon_2}} \\ &= \sqrt{\frac{\mu_2}{\epsilon_2}} \cos \theta_t = \frac{\eta_0}{\sqrt{\epsilon_{r2}}} \cos \theta_t \\ &= \frac{376.7303}{\sqrt{2}} \cos(20.70^\circ) \\ &= 249.8 \text{ } [\Omega] \end{aligned}$$

$$\begin{aligned} Z_1^{\text{TM}} &= \eta_0 \cos \theta_i \\ &= 327.1 \text{ } [\Omega] \end{aligned}$$

$$\begin{aligned} \Gamma_{\text{TM}} &= \frac{Z_2^{\text{TM}} - Z_1^{\text{TM}}}{Z_2^{\text{TM}} + Z_1^{\text{TM}}} \\ T_{\text{TM}} &= 1 + \Gamma_{\text{TM}} \end{aligned}$$

$$\Gamma_{\text{TM}} = -0.1339$$

$$T_{\text{TM}} = 0.8661$$

Note that these values do not depend on frequency for a lossless material!



Example (cont.)

Summary for TM_z Case

$$\% \text{ power reflected} = 100 |\Gamma|^2$$

$$\% \text{ power transmitted} = 100(1 - |\Gamma|^2)$$

$$\Gamma^{TM} = -0.1339$$

$$\% \text{ power reflected} = 1.79$$

$$\% \text{ power transmitted} = 98.21$$

Example (cont.)

Next, look at the TE_z case: $\Gamma_{\text{TE}} = \frac{Z_2^{\text{TE}} - Z_1^{\text{TE}}}{Z_2^{\text{TE}} + Z_1^{\text{TE}}}$

$$\begin{aligned} Z_2^{\text{TE}} &= \frac{\omega\mu_2}{k_{zt}} = \frac{\cancel{\omega\mu_2}}{\cancel{\omega}\sqrt{\mu_2\epsilon_2}\cos\theta_t} \\ &= \sqrt{\frac{\mu_2}{\epsilon_2}} \frac{1}{\cos\theta_t} = \frac{\eta_0}{\sqrt{\epsilon_{r2}}} \sec\theta_t \\ &= 285.5 \text{ } [\Omega] \end{aligned}$$

$$\begin{aligned} \Gamma_{\text{TE}} &= \frac{Z_2^{\text{TE}} - Z_1^{\text{TE}}}{Z_2^{\text{TE}} + Z_1^{\text{TE}}} \\ T_{\text{TE}} &= 1 + \Gamma_{\text{TE}} \end{aligned}$$

$$\begin{aligned} Z_1^{\text{TE}} &= \eta_0 \sec\theta_i \\ &= 436.2 \text{ } [\Omega] \end{aligned}$$

$$\Gamma_{\text{TE}} = -0.2088$$

$$T_{\text{TE}} = 0.7912$$

Note that these values do not depend on frequency for a lossless material!



Example (cont.)

Summary for TE_z Case

$$\% \text{ power reflected} = 100 |\Gamma|^2$$

$$\% \text{ power transmitted} = 100 (1 - |\Gamma|^2)$$

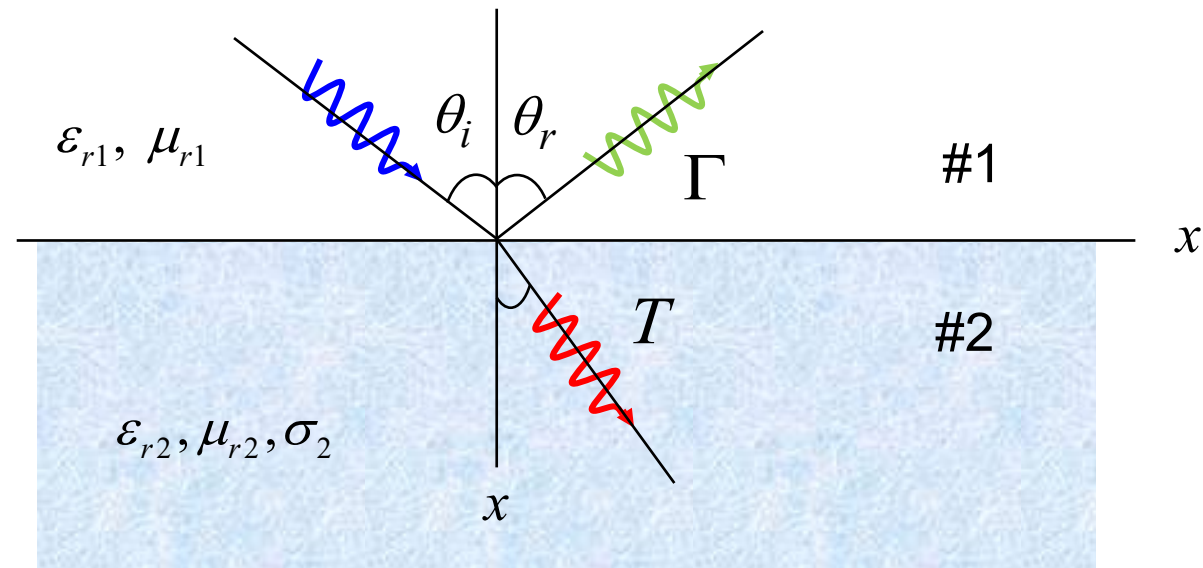
$$\Gamma^{\text{TE}} = -0.2088$$

$$\% \text{ power reflected} = 4.36$$

$$\% \text{ power transmitted} = 95.64$$

Lower region is Lossy

The lower region is now lossy.



$$\epsilon_{c2} = \epsilon_2 - j \frac{\sigma_2}{\omega}$$



$$\epsilon_{rc2} = \epsilon_{r2} - j \frac{\sigma_2}{\omega \epsilon_0}$$

Lower region is Lossy (cont.)

$$\begin{aligned}
 Z_2^{\text{TM}} &= \frac{k_{zt}}{\omega \epsilon_{c2}} = \frac{\sqrt{k_2^2 - k_{xt}^2}}{\omega \epsilon_{c2}} = \frac{\sqrt{k_2^2 - k_{xi}^2}}{\omega \epsilon_{c2}} = \frac{\sqrt{k_2^2 - k_1^2 \sin^2 \theta_i}}{\omega \epsilon_{c2}} \\
 &= \frac{\sqrt{\omega^2 \mu_2 \epsilon_{c2} - \omega^2 \mu_1 \epsilon_1 \sin^2 \theta_i}}{\omega \epsilon_{c2}} \\
 &= \frac{\sqrt{\mu_0 \epsilon_{c2} - \mu_0 \epsilon_1 \sin^2 \theta_i}}{\epsilon_{c2}} \\
 &= \eta_0 \frac{\sqrt{\epsilon_{rc2} - \epsilon_{r1} \sin^2 \theta_i}}{\epsilon_{rc2}} \\
 &= \eta_0 \frac{\sqrt{\epsilon_{rc2} - \sin^2 \theta_i}}{\epsilon_{rc2}}
 \end{aligned}$$

$$Z_2^{\text{TM}} = \eta_0 \frac{\sqrt{\epsilon_{rc2} - \sin^2 \theta_i}}{\epsilon_{rc2}}$$

$$\begin{aligned}
 Z_2^{\text{TE}} &= \frac{\omega \mu_2}{k_{zt}} = \frac{\omega \mu_2}{\sqrt{k_2^2 - k_{xt}^2}} = \frac{\omega \mu_2}{\sqrt{k_2^2 - k_{xi}^2}} = \frac{\omega \mu_2}{\sqrt{k_2^2 - k_1^2 \sin^2 \theta_i}} \\
 &= \frac{\omega \mu_2}{\sqrt{\omega^2 \mu_2 \epsilon_{c2} - \omega^2 \mu_1 \epsilon_1 \sin^2 \theta_i}} \\
 &= \frac{\mu_0}{\sqrt{\mu_0 \epsilon_{c2} - \mu_0 \epsilon_1 \sin^2 \theta_i}} \\
 &= \eta_0 \frac{1}{\sqrt{\epsilon_{rc2} - \epsilon_{r1} \sin^2 \theta_i}} \\
 &= \eta_0 \frac{1}{\sqrt{\epsilon_{rc2} - \sin^2 \theta_i}}
 \end{aligned}$$

$$Z_2^{\text{TE}} = \eta_0 \frac{1}{\sqrt{\epsilon_{rc2} - \sin^2 \theta_i}}$$

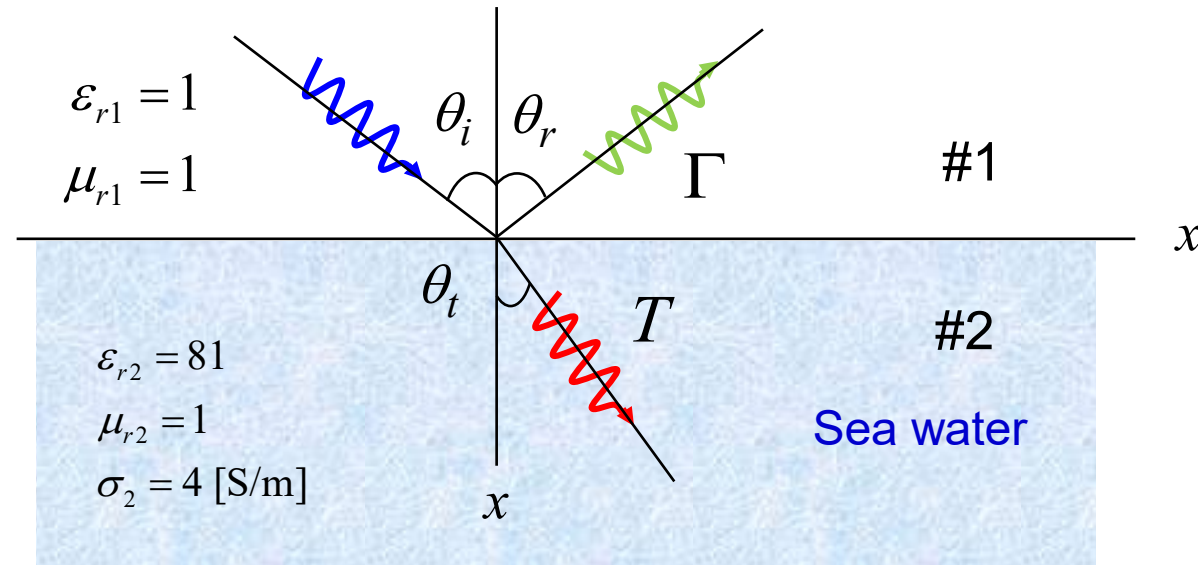
Example

The lower region is now lossy.

Given:

$$\theta_i = 30^\circ$$

$$f = 1 \text{ GHz}$$



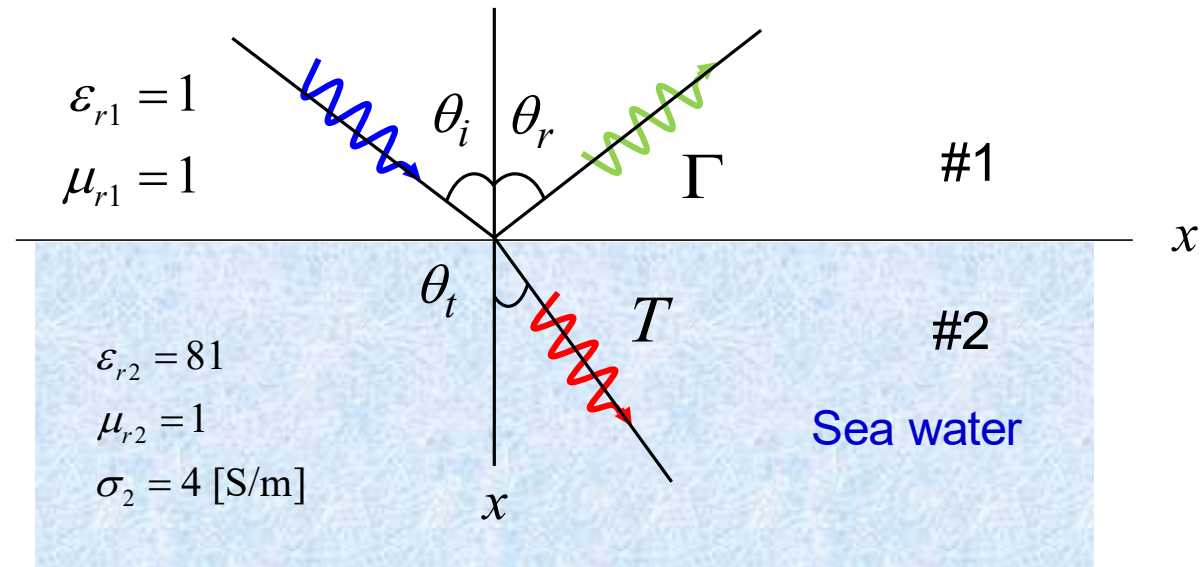
Find:

% power reflected and transmitted for a TM_z wave

% power reflected and transmitted for a TE_z wave

Example (cont.)

Given:
 $\theta_i = 30^\circ$
 $f = 1 \text{ GHz}$



$$\epsilon_{rc2} = \epsilon_{r2} - j \frac{\sigma_2}{\omega \epsilon_0}$$

$$\epsilon_{rc2} = 81 - j(71.90)$$

Example (cont.)

TM_z case:

$$\Gamma_{\text{TM}} = \frac{Z_2^{\text{TM}} - Z_1^{\text{TM}}}{Z_2^{\text{TM}} + Z_1^{\text{TM}}}$$

$$Z_1^{\text{TM}} = \eta_0 \cos \theta_i$$

$$Z_2^{\text{TM}} = \eta_0 \frac{\sqrt{\epsilon_{rc2} - \sin^2 \theta_i}}{\epsilon_{rc2}}$$

$$Z_1^{\text{TM}} = 326.3 \text{ } [\Omega]$$

$$Z_2^{\text{TM}} = 33.82 + j(12.82) \text{ } [\Omega]$$

We then have:

$$\Gamma_{\text{TM}} = -0.8099 + j(0.0644)$$

Example (cont.)

Summary for TM_z Case

$$\% \text{ power reflected} = 100 |\Gamma|^2$$

$$\% \text{ power transmitted} = 100 (1 - |\Gamma|^2)$$

$$\Gamma^{TM} = -0.8099 + j(0.0644)$$

$$\% \text{ power reflected} = 66.0$$

$$\% \text{ power transmitted} = 34.0$$

Example (cont.)

TE_z case:

$$\Gamma_{\text{TE}} = \frac{Z_2^{\text{TE}} - Z_1^{\text{TE}}}{Z_2^{\text{TE}} + Z_1^{\text{TE}}}$$

$$Z_1^{\text{TE}} = 435.0 \, [\Omega]$$

$$Z_2^{\text{TE}} = 33.86 + j(12.89)$$

$$Z_1^{\text{TE}} = \eta_0 \sec \theta_i$$

$$Z_2^{\text{TE}} = \eta_0 \frac{1}{\sqrt{\epsilon_{rc2} - \sin^2 \theta_i}}$$

We then have:

$$\Gamma_{\text{TE}} = -0.8542 + j(0.0510)$$

Example (cont.)

Summary for TE_z Case

$$\% \text{ power reflected} = 100 |\Gamma|^2$$

$$\% \text{ power transmitted} = 100 (1 - |\Gamma|^2)$$

$$\Gamma^{\text{TE}} = -0.8542 + j(0.0510)$$

$$\% \text{ power reflected} = 73.2$$

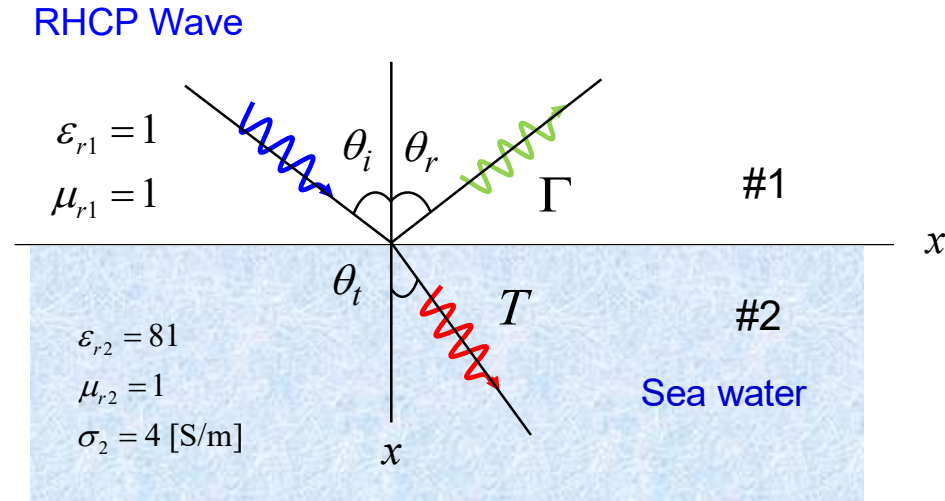
$$\% \text{ power transmitted} = 26.8$$

Example (cont.)

Now consider an incident RHCP Wave.

Given:

$$\theta_i = 30^\circ$$
$$f = 1 \text{ GHz}$$



Assume incident power density is 1 [W/m²].

$$P_i^{\text{TM}} = P_i^{\text{TE}} = 0.5 \left[\text{W/m}^2 \right]$$

$$P_r^{\text{TM}} = P_i^{\text{TM}} |\Gamma_{\text{TM}}|^2 = 0.5 \left| -0.8099 + j(0.0644) \right|^2 = 0.3300 \left[\text{W/m}^2 \right]$$

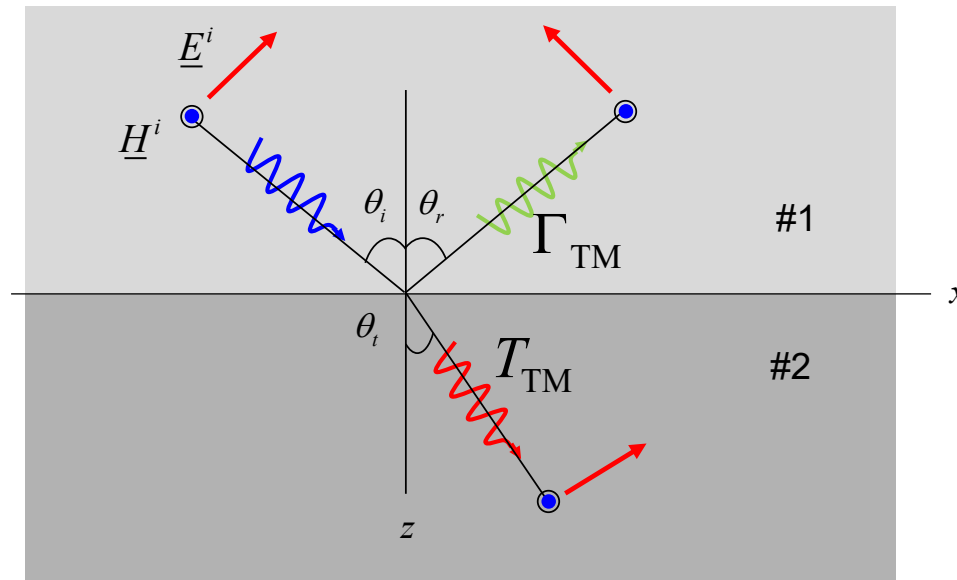
$$P_r^{\text{TE}} = P_i^{\text{TE}} |\Gamma_{\text{TE}}|^2 = 0.5 \left| -0.8542 + j(0.0510) \right|^2 = 0.3661 \left[\text{W/m}^2 \right]$$

- The reflected power is 69.6% of the incident power. ($0.3300 + 0.3661 = 0.6961$)
- The reflected wave has 47.4% of its power in the TM polarization. ($0.3300 / 0.6961 = 0.474$)

Brewster Angle

Assume TM_z polarization

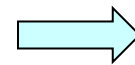
Assume lossless regions



$$\Gamma_{\text{TM}} = \frac{Z_2^{\text{TM}} - Z_1^{\text{TM}}}{Z_2^{\text{TM}} + Z_1^{\text{TM}}}$$

Set

$$\Gamma_{\text{TM}} = 0$$



$$Z_2^{\text{TM}} = Z_1^{\text{TM}}$$

Matched load!

Brewster Angle (cont.)

$$Z_2^{\text{TM}} = Z_1^{\text{TM}} \quad Z_1^{\text{TM}} = \left(\frac{k_{z1}}{\omega \varepsilon_1} \right) = \left(\frac{k_{zi}}{\omega \varepsilon_1} \right) \quad Z_2^{\text{TM}} = \left(\frac{k_{z2}}{\omega \varepsilon_2} \right) = \left(\frac{k_{zt}}{\omega \varepsilon_2} \right)$$

Hence, we have:

$$\frac{k_{zi}}{\cancel{\omega \varepsilon_1}} = \frac{k_{zt}}{\cancel{\omega \varepsilon_2}} \quad \left(k_{zt} = k_2 \cos \theta_t = \sqrt{k_2^2 - k_{xt}^2} = \sqrt{k_2^2 - k_{xi}^2} = \sqrt{k_2^2 - k_1^2 \sin^2 \theta_i} \right)$$

$$\Rightarrow \frac{k_1 \cos \theta_i}{\varepsilon_1} = \frac{\sqrt{k_2^2 - k_1^2 \sin^2 \theta_i}}{\varepsilon_2}$$

$$\Rightarrow \frac{k_1^2 (1 - \sin^2 \theta_i)}{\varepsilon_1^2} = \frac{k_2^2 - k_1^2 \sin^2 \theta_i}{\varepsilon_2^2}$$

Brewster Angle (cont.)

$$\frac{k_1^2 (1 - \sin^2 \theta_i)}{\epsilon_1^2} = \frac{k_2^2 - k_1^2 \sin^2 \theta_i}{\epsilon_2^2}$$

Assume $\mu_1 = \mu_2$:

$$\frac{\cancel{\omega^2} \cancel{\mu_1} \epsilon_1 (1 - \sin^2 \theta_i)}{\epsilon_1^2} = \frac{\cancel{\omega^2} \cancel{\mu_2} \epsilon_2 - \cancel{\omega^2} \cancel{\mu_1} \epsilon_1 \sin^2 \theta_i}{\epsilon_2^2} \quad \Rightarrow \quad \frac{1 - \sin^2 \theta_i}{\epsilon_1} = \frac{\epsilon_2 - \epsilon_1 \sin^2 \theta_i}{\epsilon_2^2}$$

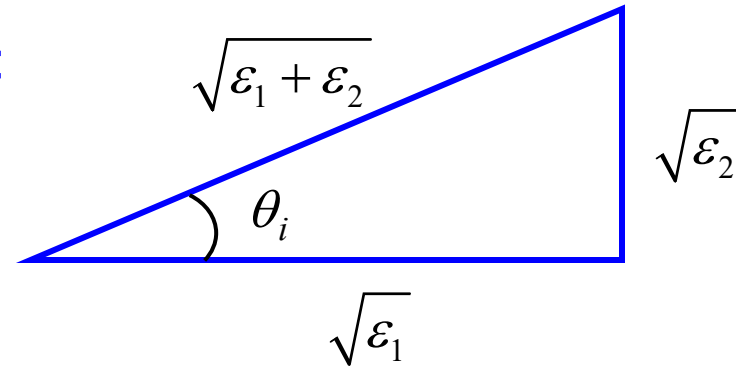
$$\Rightarrow \quad \epsilon_2^2 - \epsilon_2^2 \sin^2 \theta_i = \epsilon_1 \epsilon_2 - \epsilon_1^2 \sin^2 \theta_i$$

$$\Rightarrow \quad \sin^2 \theta_i = \frac{\epsilon_2^2 - \epsilon_2 \epsilon_1}{\epsilon_2^2 - \epsilon_1^2} = \frac{\epsilon_2 (\epsilon_2 - \epsilon_1)}{\epsilon_2^2 - \epsilon_1^2} = \frac{\epsilon_2 (\epsilon_2 - \epsilon_1)}{(\epsilon_2 - \epsilon_1)(\epsilon_2 + \epsilon_1)} = \frac{\epsilon_2}{\epsilon_1 + \epsilon_2}$$

Brewster Angle (cont.)

$$\sin \theta_i = \sqrt{\frac{\epsilon_2}{\epsilon_1 + \epsilon_2}}$$

Geometrical angle picture:



Hence

$$\theta_i = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

Brewster Angle (cont.)

This special incident angle is called the Brewster angle θ_b .

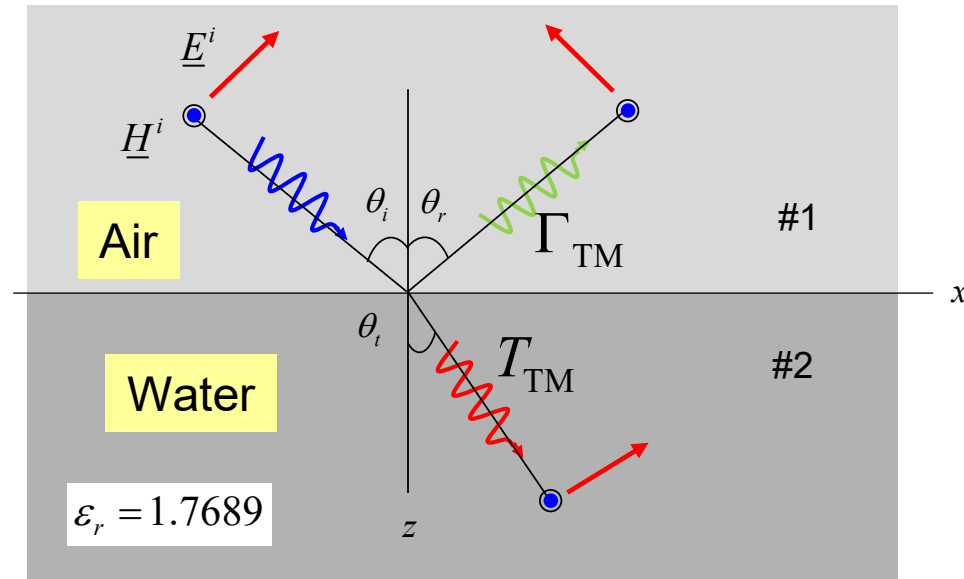
$$\theta_i = \theta_b = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

- For non-magnetic media ($\mu = \mu_0$), only the TM_z polarization (parallel polarization) has a Brewster angle.
- A Brewster angle exists for any material contrast ratio (it doesn't matter which side is optically denser).

Brewster Angle (cont.)

Example

Light goes from
air to water.

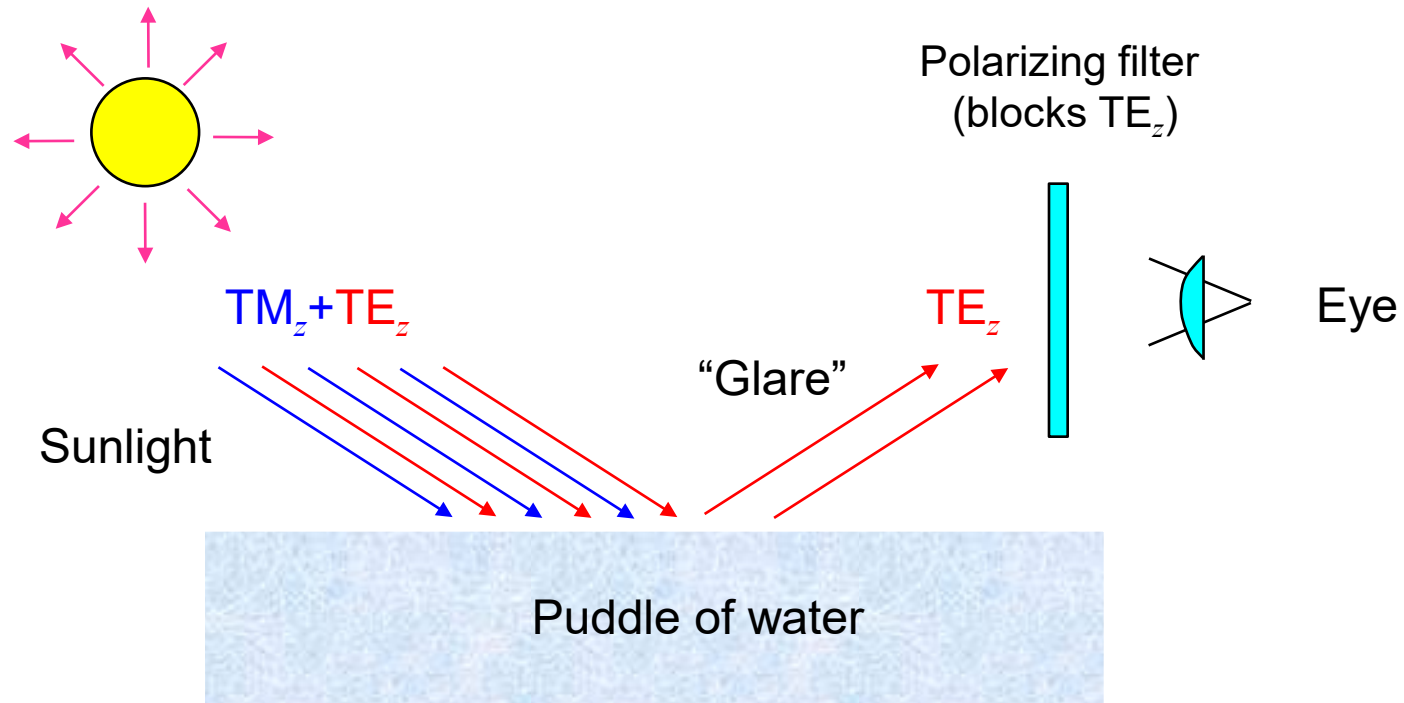


$$\theta_i = \theta_b = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}} = \tan^{-1} \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} = \tan^{-1} \sqrt{1.7689}$$

$$\theta_b = 53.06^\circ$$

Brewster Angle (cont.)

Polaroid Sunglasses



The reflection from the puddle of water (the "glare") is reduced.

Polarization Effects



A building seen through polaroid sunglasses

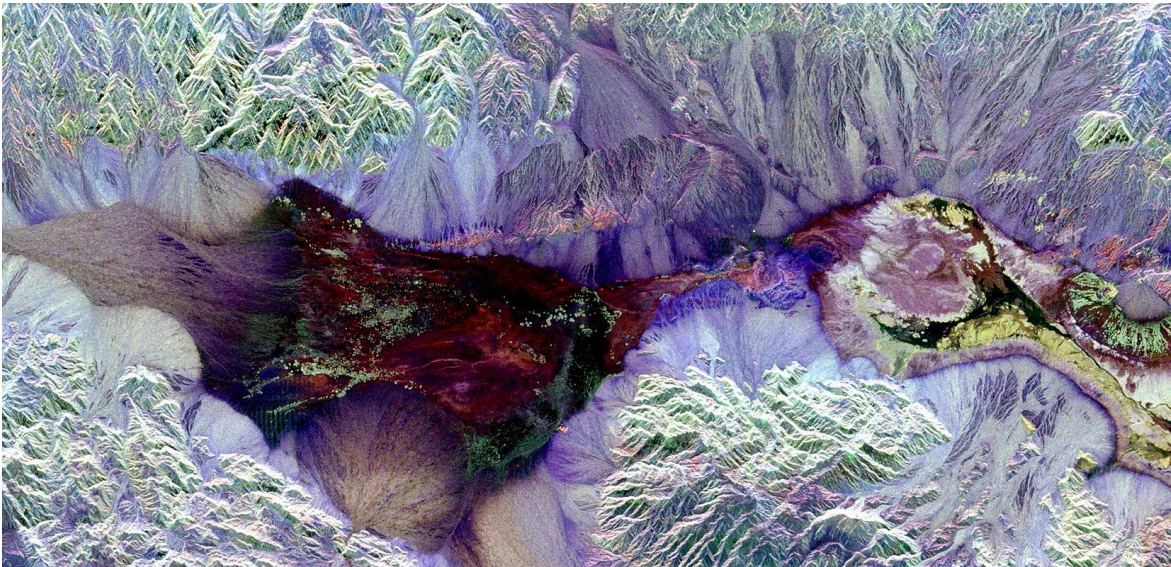
[https://en.wikipedia.org/wiki/Polaroid_\(polarizer\)](https://en.wikipedia.org/wiki/Polaroid_(polarizer))

By Jason7825 at English Wikipedia

Polarization Effects (cont.)

“**Polarimetry** is the measurement and interpretation of the polarization of transverse waves, most notably electromagnetic waves, such as radio or light waves. Typically polarimetry is done on electromagnetic waves that have traveled through or have been reflected, refracted or diffracted by some material in order to characterize that object.”

<https://en.wikipedia.org/wiki/Polarimetry>



Synthetic aperture radar
image of Death Valley
colored using polarimetry.